Combinatorial Encoding of Visual Contrast into Gamma Rhythm Features

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Visual cortical activity elicited by a static grating exhibits a sharp broadband transient that settles into a highly variable gamma rhythm (30-90 Hz). The operating regimes underlying such neural responses, and the possibility of transitions between brain states, are important open questions. Taking advantage of a recent simple envelope-phase description of network gamma rhythms, we fit local field potential data from macaque area V1 to two meta-parameters that represent effective stability and noise. This enables pinpointing the dynamical regime before and during stimulation. The neuronal noise in this efficient model accounts for the random bursts of gamma power and trial-to-trial variability. This framework establishes how a low-to-moderate stimulus causes cortical state changes from a noisy non-oscillatory equilibrium to a noise-driven focus attractor (a quasi-cycle) with high Fisher information. Gamma bursts arise in an Inhibition-Stabilized Network (ISN) regime with near-zero net synaptic currents. Strong stimuli move the system beyond the ISN, resulting in weaker gamma responses as seen in the data. Our analysis explains transients following stimulus onset and the correlation of stimulus strength with burst duration, network frequency, firing rates and effective synaptic coefficients. Different contrasts elicit varying degrees of normal and non-normal amplification with associated changes in burst statistics. Our results highlight a combinatorial encoding where each contrast elicits a unique combination of network frequency, stability and noise with specific gamma burst statistics.

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The encoding of stimulus features into spatially and temporally varying patterns of cortical brain activity remains an active area of research with a number of open questions. In particular, the seemingly simple task of looking at a periodic array of black and white bars induces a fluctuating oscillatory response in local field potentials (LFPs) of the primary visual cortex (area V1), in both non-human and human primates [1, 2]. Yet the nature of the dynamical brain states before and during the stimulus that allows such complex responses is still elusive, as is the system's level strategy to represent information about the visual world into the observed noisy responses. Indeed, after a short transient of high power over a broad (30-90 Hz) frequency range, these induced "gamma" rhythms exhibit high variability, both during trials and across trials in response to the exact same stimulus. The rhythms manifest as short and random epochs of synchrony known as "gamma bursts". The duration, amplitude and frequency content of the bursts are random variables [3–5] that somehow encode - or at the very least correlate with - stimulus features, and make that information available to deeper visual processing centers in spite of the variability. Bursts are in fact seen across many rhythms where they raise similar questions (Lundqvist2018).

Beyond coding questions, we seek insight into which oscillatory mechanism is engaged when a stimulus such as a grating appears. The dynamical regimes that enable stimulus-induced gamma rhythms are not well understood, a question made difficult by response variability. Do detailed network properties matter, or can one reason - as we do here - with simple concepts of effective excitation and inhibition? Is nonlinearity even necessary, since noise linearizes network dynamics (Brunel 2000,Doiron 2003)? The resulting transition from noisy quiescence to noisy oscillation may lead to changes in the frequency and stability of the neural circuit dynamics, but also in the level of noise in which they operate. These three characteristics are shown here to be key to understanding the mechanism for representing inputs into rhythms. Our paper presents a method for quantifying frequency, stability and noise, and consequently for pinpointing the necessary operating regime to perform encoding and communication.

Specifically, we develop a joint modeling, parameter inference and Fisher information framework that addresses head-on the plausible dynamical nature of the LFP fluctuations recorded from monkey V1 as it processes grating onsets and offsets. It allows a probabilistic description of optimal stimulus representation using rhythm bursts driven by neuronal noise, and thus highlights a novel important role for this ubiquitous noise. Our work focuses on gamma rhythms in V1 being generated by interacting excitatory (E) and inhibitory (I) neurons (the "PING" mechanism) when the stimulus is presented [6]. Further, rather than invoking deterministic oscillators jittered by noise, our work is predicated on the recently promoted alternate view of a stimulus-driven stable focus attractor [3, 7–9]. In this dynamical picture, noise induces fluctuating oscillations, or "quasi-cycles", that mimic the random epochs (i.e. bursts) of heightened oscillatory power seen experimentally during stimulation. This enables us to state the nature of the stimulus-free regime, and the optimal encoding capabilities of stimulus intensity (here, grating contrast) into the features of gamma bursts, a methodology applicable to other bursty rhythms.

Rhythm burst are unavoidable features of rhythm variability, yet their dynamical properties and putative functions are far from understood. Recent studies have linked them to phase synchronization [10], flexible interareal brain communication [11, 12], as well as stimulus encoding-decoding and working memory [13–15], for which their longer duration is thought to be favorable. Dysfunction in stimulus amplification manifested in increased burstiness has been linked to neurological disorders; for example, mean burst amplitude and duration are reduced in Alzheimer's disease

[16]. By further developing our previous envelope-phase reduction work based on the stochastic averaging method to the context of externally-driven systems, here we arrive at a novel relation between burst dynamics and stimulus properties, which our fitted models can account for. Upon increasing stimulus contrast, a surprising path involving varying mixtures of so-called normal and non-normal amplification is seen in an inhibition-stabilized regime. These explain observed gamma burst features, and can account for both monotonic and non-monotonic mappings of contrast into rhythm spectral power and information content seen experimentally both at stimulus onset and in the steady state.

I. RESULTS

Envelope-phase description of stimulus-driven LFPs. The V1 circuitry is modeled simply as effective self- and cross-interactions of its E and I neuron populations (see Methods and Fig.1**A**). The E-I dynamics consist of two linear coupled stochastic differential equations [17] obtained from the linearization of the nonlinear Wilson-Cowan model [18, 19]; these generic dynamics also result from other formulations of two-dimensional excitatory-inhibitory dynamics. This class of dynamics is chosen based on its known ability to autonomously exhibit LFP-like oscillations of varying regularity depending on parameters. The steady state response to a grating stimulus is considered at first; the transient response at stimulus onset is addressed afterwards. As our analysis will demonstrate, a plausible operational regime for the dynamics is one where gamma activity relies on both the stimulus and neuronal noise [3]. The randomly occurring epochs or "bursts" of higher gamma power can be delineated by focusing on the slow envelope of the LFP amplitude. An envelope-phase description of the E-I activities in this noise-induced oscillatory regime has been derived [7, 9] (see Methods). This includes an expression for the probability density of envelope values (Methods). Specifically, the gamma LFP is assumed to follow [9]:

$$V(t) = Z(t)\cos\big(\omega_0 t + \phi(t)\big)\,. \tag{1}$$

The envelope $Z(t)$ and the phase $\phi(t)$ evolve according to:

$$dZ(t) = \left[-\lambda_0 Z(t) + \frac{D}{2Z(t)}\right]dt + \sqrt{D}dW_1(t) \tag{2}$$

$$d\phi(t) = \frac{\sqrt{D}}{Z(t)}dW_2(t)\,, \tag{3}$$

where $W_1(t)$ and $W_2(t)$ are two independent Brownian motions (their time derivatives are Gaussian white noises). The envelope process is uncoupled from the phase process. Its stationary probability density function (pdf) is:

$$P(Z) = \theta Z \exp\left(-\theta\frac{Z^2}{2}\right). \tag{4}$$

This envelope density depends only on the meta-parameter

$$\theta = \frac{2\lambda_0}{D}\,. \tag{5}$$

The coefficient λ_0 is the absolute value of the real part of the complex conjugate eigenvalues of the associated E-I network. The meta-parameter is the ratio of this stability coefficient (Methods, Eq.44: loss of linear stability occurs for $\lambda_0 \to 0$) to an effective total noise strength driving the E and I dynamics (Eq.53). The envelope value $Z = Z_{max} = \sqrt{1/\theta}$ maximizes the pdf $P(Z)$. The parameters λ_o and D are functions of all the original parameters of the stochastic Wilson-Cowan (SWC) system. Our analysis below exploits this strong reduction in number of parameters. This reduction also appears in the envelope dynamics Eq.2, which can be further non-dimensionalized to show a dependence only on θ [20]. Additionally, the parameter ω_0 (Eq.45) is necessary to describe the LFP mean frequency. The envelope-phase framework allows a joint dynamical and probabilistic description of the envelope of the LFP, and in particular of the bursts, whose features can be fitted to the data given their simplicity. Finally, the normalized autocorrelation function of the LFP can be derived (see Methods, Eq.1) as

$$C(\tau) = \big[\cos(\omega_0\tau) + C_0 \sin(\omega_0\tau)\big] \exp(-\lambda_0\tau)\,. \tag{6}$$

Envelope coding and Maximum Likelihood Inference. We assume that information about the stimulus contrast is encoded in the gamma rhythm envelope. This envelope coding intuition rests on the fact that the parameter θ in the envelope pdf contains all the parameters of the model. Decoding of the contrast can then be assessed via the likelihood that a given stimulus gave rise to the observed sequence of envelope values; this likelihood depends only on θ. For each contrast, we extract the envelope process through the analytic signal values (the norm obtained using the process and its Hilbert transform) $\tilde{Z}_j = \tilde{Z}(t_j)$ of each data set $\tilde{\mathbf{Z}}$. We consider that the information about the stimulus contrast is contained in the approximately independent successive values $\tilde{Z}_j, j = 1, ..., n$ of the envelope process $\tilde{\mathbf{Z}}$. Using the envelope density in eq.4, the likelihood becomes (see Methods):

$$L(\theta) = \prod_{j=1}^{n} \theta \tilde{Z}_j \exp\left(-\theta \frac{\tilde{Z}_j^2}{2}\right).$$

We estimate the value $\hat{\theta}$ of θ that maximizes the likelihood function $L(\theta)$ (see Methods and Fig.1**B-Left**):

$$\begin{aligned} \hat{\theta} &= \arg\max_{\theta}\ \log L(\theta) \\ &= \frac{2}{\langle \tilde{\mathbf{Z}}^2 \rangle}\,. \end{aligned} \tag{7}$$

Thus for each stimulus contrast, a maximal likelihood estimate (MLE) of θ is obtained. We also compare the probability density functions computed as histograms of the analytic signal values of the envelope process $\tilde{\mathbf{Z}}$ with the theoretical expression Eq.4 in which θ is replaced by its maximum likelihood estimate Eq.7). Good agreement is found between the experimental and inferred densities Fig.1**C**. However, probably because the length of our datasets was relatively short, some estimates were corrected to obtain better values.

To further quantify the information that the envelope of the gamma LFP carries about the stimulus contrast, we

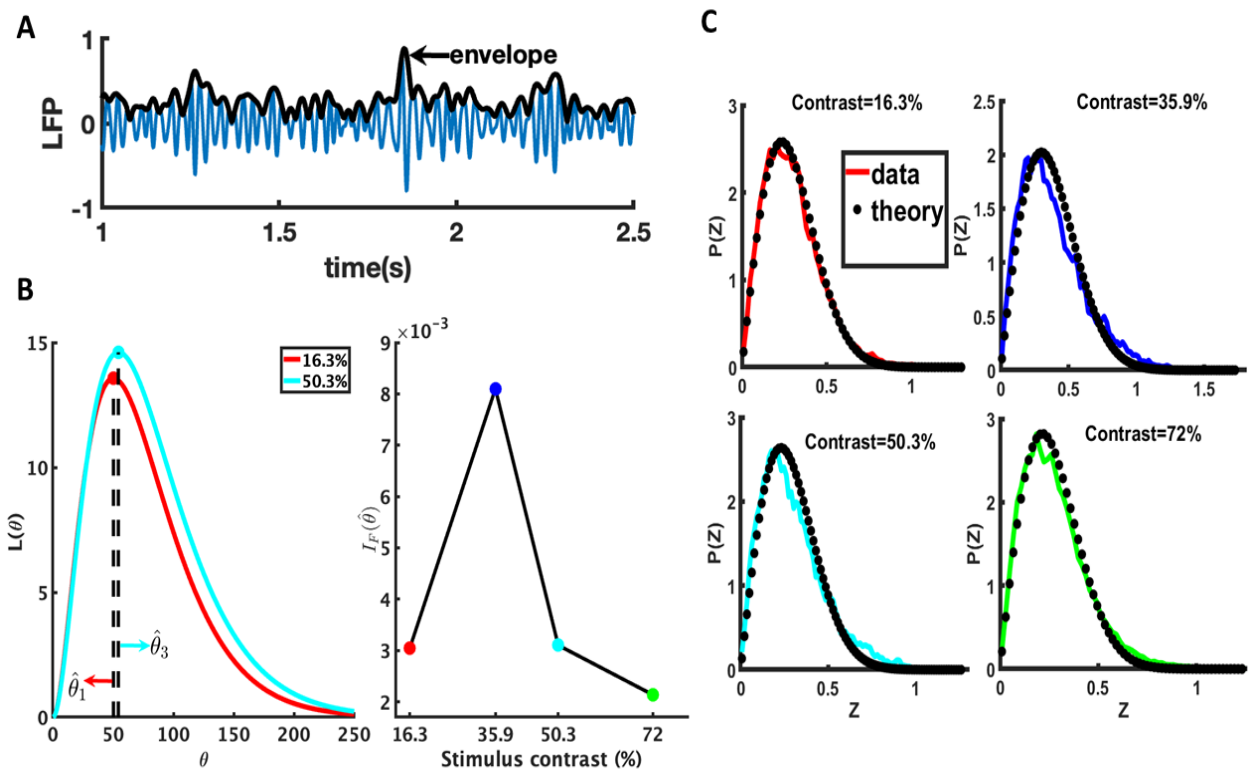

FIG. 1. **Gamma rhythm envelope encoding of visual contrast in monkey V1. A:** Gamma rhythm response (blue) and associated envelope (black) during a single trial for a stimulus contrast of 16.3%. **B-Left:** Example of likelihood curves for two stimulus contrasts; their peaks correspond to Maximum Likelihood Estimates of the parameter θ. **B-Right:** Fisher Information as a function of stimulus contrast. **C:** Probability density function for the envelope for each contrast computed as normalized histograms of the analytic signal representation of the data via the Hilbert transform (red, blue, cyan and green) compared to the theory using the parameters estimated using MLE (black).

compute the Fisher information (FI) as (see Methods):

$$I_F(\hat{\theta}) = \frac{n\left(\langle \tilde{\mathbf{Z}}^2 \rangle\right)^2}{4}\,.$$

The Fisher information depends on the second moment of the envelope amplitude $\langle \tilde{\mathbf{Z}}^2 \rangle$, which we refer to as the amplification. The stimulus with 35.9 percent contrast has the highest Fisher information, while the stimulus with 72 percent contrast has the lowest (Fig.1**B-Right**). A higher Fisher information means that contrast changes around this contrast value cause the strongest variations in the parameter θ relative to its fluctuations, such that θ then most sensitively encodes the contrast. The stimuli with 16.3 and 50.4 percent contrasts have moderate FI, lying between those for the 35.9 and 72 percent stimuli. Therefore, stimulus contrast has a non-monotonic effect on the envelope amplification with a sharp increase around moderate values.

Data-driven reconstruction of envelope-phase model. Some information about stimuli can be obtained from gamma rhythm envelopes through the MLE. However, these estimates do not by themselves allow the reconstruction of the envelope-phase dynamics. To this end, we must infer the parameters λ_0 and ω_0 from the LFPs and deduce the parameter D from the envelope MLE. We show (see **Methods**) that λ_0 is the decay rate of the normalized

autocorrelation $C(\tau)$ of the LFPs. The autocovariance $C(\tau)$ is an even oscillatory function, and its peaks exhibit a decay of the form $\exp(-\lambda_0\tau)$. The value of λ_0 can be inferred from $C(\tau)$ (see Methods, Eq.44). The parameter $\omega_0 = \frac{2\pi}{T_0}$ corresponds to the rhythm's mean angular velocity. The mean period T_0 is estimated from the distance between the central peak and the neighboring local maximum in the autocorrelation.

If we take into account the expression of the envelope MLE Eq.7 and the expression of θ from the envelope pdf Eq.5, we can deduce:

$$D = \lambda_0 \langle \tilde{Z}^2 \rangle \,. \tag{8}$$

The envelope-phase equations 2-3 can be completely reconstructed from the inferred parameters λ_0, ω_0 and D. We perform the same inference procedure for all contrasts. The mean frequency increases with contrast (Fig.2**B-right panel**) as in the data, consistent with previous studies [1, 21]. The relations between contrast and the parameters λ_0 and D are non-trivial (Fig.2**B, left and middle panels)**, unlike for ω_0. The stability parameter λ_0 is relatively constant except for the strongest contrast, while the effective noise D varies non-monotonically with contrast. One can also see that each contrast is encoded by a unique combination of these three parameters, i.e. **contrast encoding is combinatorial**.

We further perform numerical simulations of the reconstructed envelope-phase models (Fig.3**B**) and compare the obtained envelope density (after transients) with those from the data (obtained via the Hilbert transform) (Fig.3**C**). We also compare the angular frequencies (Fig.3**D**) and the whole LFP probability distributions (Fig.3**E,F**). For each comparison, excellent agreement is found between the reconstructed envelope-phase models and recorded LFPs. The comparisons in Fig.3 are for the 16.3% contrast. Parameter fitting results for the other contrasts can be found in Fig.S1, while the correspondence between the fitted envelope-phase model envelope densities and time series spectrograms and joint distributions are shown in Figs.S2, S3 and S4 for the other contrasts.

Together these results establish the validity and consistency of our method, and further reveal the one-to-one encoding of stimulus contrast in a unique combination of gamma rhythm features. However, these analyses do not tell us about the similarity of bursting characteristics in the steady state LFP activity.

Burst feature analysis. The time-series are scanned to detect all the bursts. The number of oscillation cycles is determined for each burst in order to classify it as True (TB) or False (FB) burst (Fig.4). For each burst in these two classes, we computed the corresponding duration, amplitude and frequency (see Methods). From this procedure, we build histograms of TB and FB durations, amplitude and frequency for the four contrasts (see Figs.S5-8), for both the experimental data and those generated from our inferred models. We compute the mean burst duration, amplitude and frequency, and their standard deviations. Comparison of data and simulations reveal excellent agreement for the four stimulus contrasts. The burst features bear the identity of the envelope process and the whole gamma rhythm response. Their properties help infer the stimulus identity and evaluate the encoding capacity of the gamma rhythms, as we now show.

Two types of Envelope Amplification. To understand the mechanisms that lead to distinct burst features

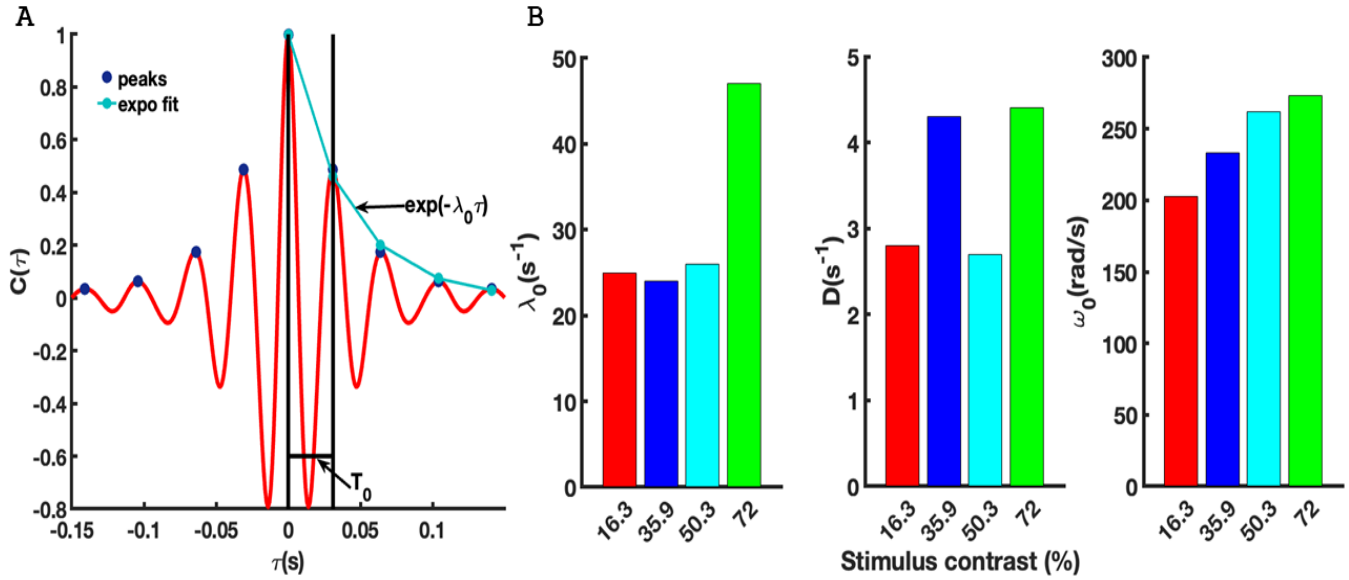

FIG. 2. **Combinatorial coding of contrast with network stability, effective noise and mean gamma frequency.** The fitting of the envelope-phase model reveals different combinations of dynamical parameters across contrasts. **A:** Autocorrelation of the LFP for a stimulus contrast of 16.3% (blue line), along with its local peaks (red circles) used to fit an exponential (green line). **B:** Fitted parameters for each stimulus contrast: the stability parameter λ_0, the effective noise D and the mean angular network frequency ω_0.

(burst amplitude and duration) induced by stimuli 1 to 4, we compute the mean envelope amplification as

$$\langle Z^2 \rangle = \int_0^\infty z^2 P(z) dz = \frac{D}{\lambda_0}. \tag{9}$$

Therefore, the envelope can be amplified by either increasing D or decreasing λ_0. Decreasing λ_0 corresponds to bringing the E-I network close to the loss of linear stability. This results in more bursts having simultaneously long duration and amplitude which we refer to as True/Long Bursts. At the transition to the linear instability ($\lambda_0 = 0$) the theory predicts an infinite amplification for the envelope process, leading to pathological states. This amplification is known as the normal amplification. Note that the theory has been developed further to reach beyond the transition [22], but its parameters are more difficult to estimate, which is beyond the scope of our work.

The coefficient D is proportional to the noise strength with a proportionality coefficient that depends on the E-I network synaptic coefficients (see Methods). Increasing D amplifies the envelope process, but in contrast to the normal amplification, it does not necessarily lead to loss of linear stability. This second type of non-normal amplification may be related to the E-I network having a hidden feedforward structure. Practically, non-normal amplification is associated to small amplification of the envelope process and should be related to the production of shorter bursts (False/Short bursts).

Our analysis reveals that both types of amplification are simultaneously present, but in a proportion that depends on the stimulus. The fitting procedure in Fig.2 yields ($\lambda_0 = 25.0634, D = 2.7682$), ($\lambda_0 = 23.8487, D = 4.2928$) , ($\lambda_0 = 24.9695, D = 2.7810$) and ($\lambda_0 = 47.3835, D = 4.3806$) for stimuli 1 to 4, i.e. from lowest to highest contrast. From these values we can quantify the proportions and interpret the burst features. (ARTHUR: CHECK THE FOLLOWING) Stimuli 1 and 3 have low values of λ_0, with a lower value for stimulus 3. The corresponding values of D are intermediate. Therefore, their respective normal amplifications are high (with a higher value for stimulus 3

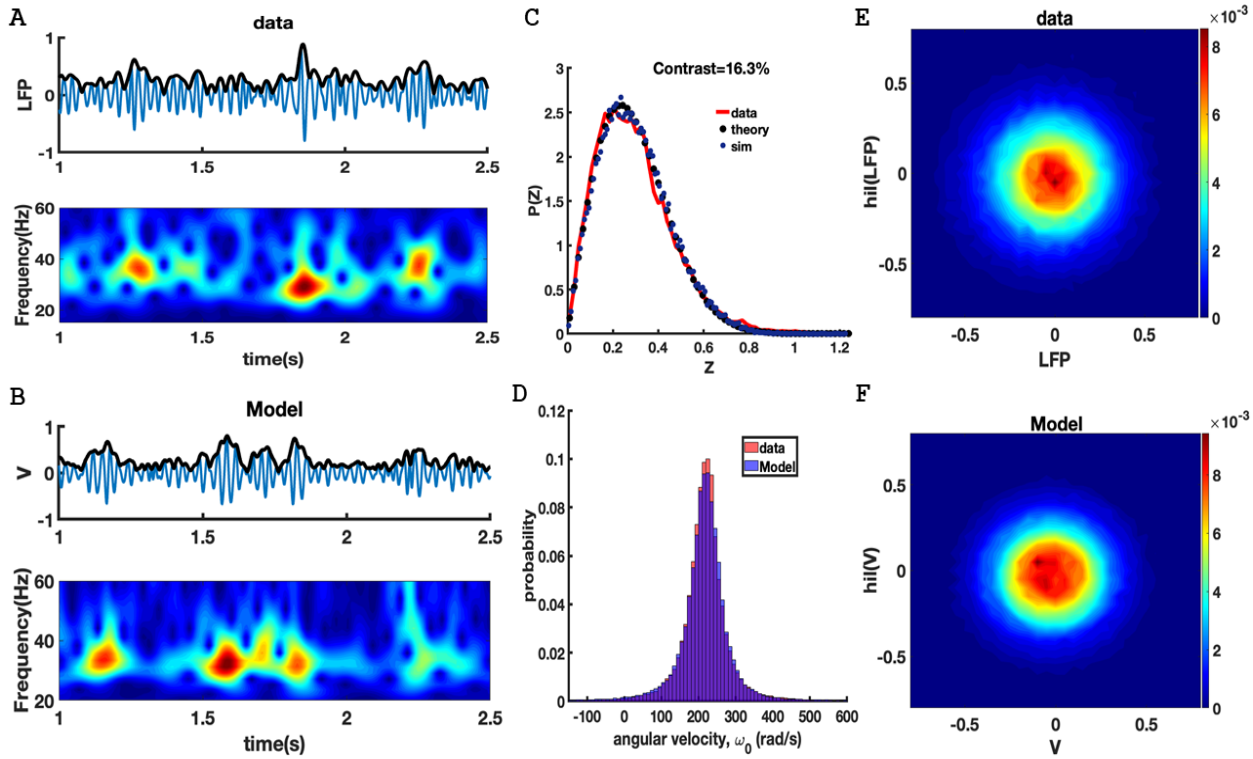

FIG. 3. **Parameter fitting of the envelope-phase model reconstructs LFP properties. A:** Measured gamma rhythm LFP (top) and the corresponding spectrogram (bottom) for a single trial at 16.3% contrast. **B:** Gamma rhythm LFP (top) and associated spectrogram (bottom) generated by the envelope-phase model after the fitting procedure. **C:** PDF of the envelope values for the data (red), the theory using the fitted parameters (black dots), and the LFPs generated by numerically integrating the envelope-phase model with the fitted parameters (blue). **D:** PDF of the angular velocity estimated from the data (red) and the numerical integration of the fitted envelope-phase model (blue). **E:** Joint distribution of the LFP data and associated analytic signal amplitude (via the Hilbert transform). **F:** Same as in **E**, but for the simulation of the fitted envelope-phase model.

due to the lower value of λ_0) and their non-normal amplifications are intermediate. This results in bursts with longer durations (130 ms $\pm$46.3094 ms and 138.5585 ms $\pm$ 62.055 ms for stimuli 1 and 3, respectively) but intermediate values of their amplitudes (0.5745 $\pm$ 0.1342 and 0.5842 $\pm$ 0.1778, respectively).

Stimulus 4 has the highest value of λ_0 resulting in the lowest normal amplification. However, the value of D is the highest. That stimulus induces the highest non-normal amplification. The corresponding envelope process produces True bursts with relatively shorter durations (96.5970 ms $\pm$ 36.5629 ms) but higher amplitudes (0.6354 $\pm$ 0.1802). Stimulus 2 results in the lowest value of λ_0 and a higher value of D (compared to stimulus 1 and 3). The normal and non-normal amplifications are both high, resulting in a total amplification higher than those for stimuli 1, 3 and 4. The envelope process for stimulus 2 has True bursts with longer durations (161.1074 ms $\pm$ 79.7227 ms) and amplitudes (0.6873 $\pm$ 0.2230). Therefore, the precise values of λ_0 and D shape the bursting features of the induced rhythms. Each stimulus shifts the E-I network to a region of parameter space that corresponds to specific values of the meta-parameters λ_0 and D.

Burst Dynamics in the (λ_0,D) parameter space. How do the two meta-parameters λ_0 and D shape the burst dynamics more generally? How do they relate to the stimulus encoding capacity of V1? Each is now varied while

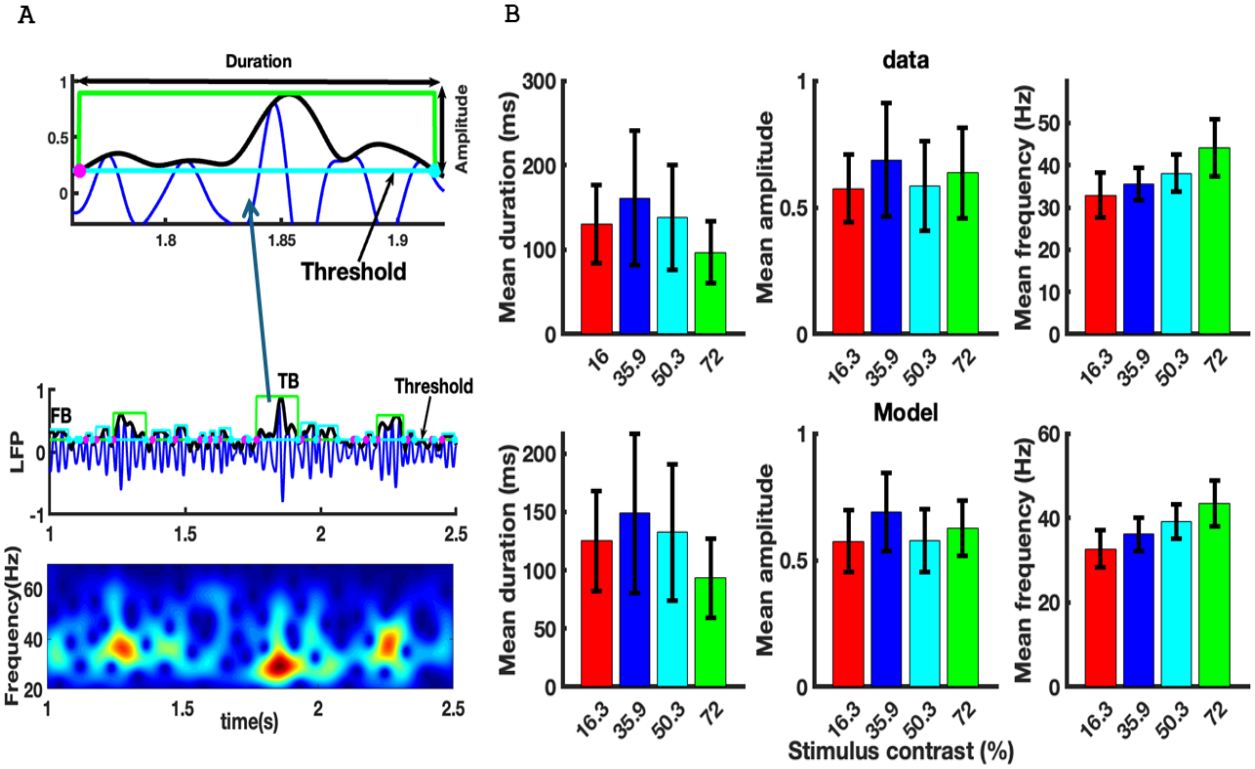

FIG. 4. **Burst definition, detection, classification and statistics. A** A burst (top) is defined as an epoch during which the black envelope (middle) of the fast gamma rhythm (blue) exceeds a chosen threshold (cyan), with associated higher gamma power as shown in warm colours in the spectrogram (bottom). The burst duration and amplitude are highlighted. In a True burst (TB), the envelope spends at least three cycles of the fast rhythm above the threshold, otherwise it is a False burst (FB). **B** Mean duration, amplitude and frequency of the True bursts corresponding to the data (top) and the fitted envelope-phase model (bottom). The error bars correspond to one standard deviation.

keeping the other fixed, and the mean duration and amplitude of the False and True bursts are computed. Increasing D increases the mean True burst duration. Decreasing λ_0 has the same effect. We also computed the mean amplitude of the True bursts for various values of D and λ_0. We found that increasing D and decreasing λ_0 both increase the mean duration of the True bursts.

We compute the Fisher Information for the same values of the parameters D and λ_0. We found that the Fisher Information is maximal for both high values of D and low values of λ_0. Qualitatively, our analysis shows that optimal stimulus encoding (high Fisher information) arises when both the True burst mean duration and amplitude are high via the combination of normal and non-normal amplifications. To complete our analysis, we also compute the mean burst duration and amplitude of the False bursts as we both vary λ_0 and D. Decreasing λ_0 increases the duration of False bursts, but not substantially, all the while keeping the False burst mean amplitude almost constant. Increasing D does not substantially increase the mean False burst duration but substantially increases the mean False burst amplitude.

(ARTHUR: THIS PARAGRAPH NEEDS WORK: REDUCE AND CLARIFY, OR GET RID OF IT- VIRTUAL BURSTS IS NOT A CONCEPT THAT HAS BEEN DEFINED. YOU CAN START YB REMOVING THE FIRST TWO SENTENCES WHICH ONLY REPEAT PREVIOUS SENTENCES) The analysis of False bursts allows to better understand the effects that the parameter λ_0 and D have on the envelope amplification. In other words, the

analysis of False bursts shows how the normal and non-normal amplification generate and shape the bursts. The fact that increasing the value of D increases the False burst mean amplitude but leaves the mean duration almost constant shows that the main effect of D and its corresponding non-normal amplification is to increase the amplitude of the envelope process by generating short transients of high amplitudes. These short transients create new False bursts where there was virtually no previous bursts or add themselves to previous virtual False or True Bursts resulting to more True bursts with higher amplitude and duration. In other side, reducing λ_0 mostly increases the duration of existing False Bursts. This increase of the duration is accompanied with a small increase of the mean False burst amplitude. Therefore λ_0 will amplify the envelope process by elongating the duration of existing bursts resulting to more True Bursts. For short, both λ_0 and D shape the burst duration and amplitude. However, the main effect of D is to shape the burst amplitude while λ_0 mostly shapes their duration.

Building corresponding data-driven E-I networks. From the three parameters extracted above, we now seek information about the connection weights W_{EE},W_{EI},W_{IE} and W_{II}, the external currents h_E and h_I, the excitatory and inhibitory decay rates α_E and α_I, the excitatory and inhibitory response function thresholds β_E and β_I, and the noise intensities σ_E and σ_I, a total of 12 parameters. A complete fitting of the 12 network parameters is a challenging problem beyond the scope of the present work; this full inference is potentially solvable with machine learning techniques, with likely a range of similar solutions that can be narrowed by experimentally known constraints.

Our goal here is instead to show how the meta-parameters extracted from the envelope analysis reveals more detailed information about the E-I network. The premise is our knowledge about which parameters bring the network near oscillatory dynamics, which requires a balance between feedback inhibition which promotes it, and recurrent excitation which impedes it (as it promotes bistability). Accordingly, we sacrifice detailed inference yet show using bifurcation theory that it is useful to focus on only two synaptic weights of expected high relevance: recurrent excitation and feedback inhibition. The inferring procedure is then divided in the two following steps.

1) We first identify the types of bifurcations in our network governed by the curves $\lambda_0\big(W_{EE},W_{EI}\big)=0$ and $\omega_0\big(W_{EE},W_{EI}\big)=0$ are the main possible bifurcations. The first bifurcation (Fig.**??A-D**, black-dotted line) separates two oscillatory regimes. On the left lies a stable oscillatory regime that requires noise to show gamma rhythms consistent with the *in vivo* data. On the right, a linearly unstable regime exhibits rhythms with diverging amplitude that are incompatible with the data. The latter unstable oscillatory regime is in fact a stable limit-cycle regime when nonlinearity is considered and bounds the divergence. Nevertheless, it will still be inconsistent with the data. The second bifurcation (Fig.**??**-Top, dark-red dotted line) distinguishes the two previous oscillatory regimes (stable and unstable) on the top and a stable non-oscillatory regime at the bottom. Plotting the bifurcation curves allows us to anticipate the operating regimes of the network.

2) We plot the curves $\lambda_0\big(W_{EE},W_{EI}\big)=\lambda_0^i$ and $\omega_0\big(W_{EE},W_{EI}\big)=\omega_0^i$ in the $\big(W_{EE},W_{EI}\big)$ plane respectively in dashed and solid lines. The intersection of these two lines is a unique inferred pair of synaptic coefficients. We repeat the same procedure for the four stimuli. The results are the colored dashed and full lines in Fig.**??A-D**, which and their corresponding intersections. Precisely, the intersections of the full and dashed red, blue, cyan and green lines refer to the first, second, third and fourth datasets, respectively (see Fig.**??-Top**). (ARTHUR: THIS FIGURE LABELING IS WRONG; IT SHOULD BE REFERRING TO FIG.5, NOT FIG.6A-D).

State-dependent values of the noise. It is possible to push our inference procedure further to determine the

actual noise intensities from the extracted meta-parameter D. We assume for simplicity that $\sigma_E{=}\sigma_I{=}\sigma$. The quantity (see Methods)

$$\frac{D}{\sigma^2} = \frac{1}{2\sin^2(\delta)}\left(1 + \frac{a}{b}\frac{W_{EI}}{W_{IE}}\right). \tag{10}$$

can be directly computed from the two inferred synaptic weights. The noise intensity is then obtained from the value of D extracted from the data. However, this procedure seems a little bit trivial and does not rely on any biophysical insight. (ARTHUR: WHY MENTION IT THEN??)

Amplification mechanisms in an E-I recurrent network. Our envelope-phase model depends on only three meta-parameters making it attractive for inference. However, it is not easy to give a biophysical interpretation to those meta-parameters. In other words, understanding how a biophysically realistic neural network can achieve the two distinct types of amplification and generate primary visual cortex data is not an easy task. How can we realize the normal and non-normal amplifications explained, in a recurrent network? What are the synaptic coefficients involved in these amplifications? To answer these questions we have derived theoretical expressions for the meta-parameters (see **Methods**). Linear stability and fitting methods of the two-dimensional E-I network allows to identify dynamical regimes of oscillations (see Fig.**??-Top**) and the corresponding eigenvalues. The coefficient λ_0 is the absolute value of the real part of the complex conjugate eigenvalues and is given by

$$\lambda_0 = -\,aW_{EE} + bW_{II} + c\,. \tag{11}$$

Furthermore, the coefficients a, b and c are all positive and depend on the E and I fixed point activities as well as on the single neuron time constant (see **Methods**). Increasing the recurrent excitation W_{EE} will eventually decrease λ_0 and lead to the normal amplification. Therefore, normal amplification can be achieved in an E-I recurrent neural network by increasing the excitatory recurrent excitation. An illustration is shown in Fig.**??-Bottom** for three constant values of the feedback inhibition. Increasing the recurrent excitation decreases λ_0 and then leads to network amplification. This also leads to an increase of the network frequency, firing rates and total synaptic inputs. The values of the parameters $\frac{D}{\sigma^2}$ remain almost constant, at least for weak and intermediate values of the feedback inhibition chosen here. This suggests that the network amplification is essentially due to the recurrent excitation getting close to the bifurcation curve in black dotted line (an illustration of the transition to the bifurcation is shown in Fig.**??-Top** with dots for constant W_{EI}). This clearly shows what normal amplification refers to in a biophysical E-I neural network.

The non-normal amplification is obtained by increasing the effective noise strength D that is proportional to the noise inputs to the E and I populations. Since we do not directly have access to these noisy inputs through the data, an appropriate noise-free measure of the non-normal amplification is $\frac{D}{\sigma^2}$. It depends on several coefficients. The coefficient γ is a positive constant that depends on the E and I fixed points and single neuron decay rate. The coefficient δ is the phase-difference between the I and E activities. For fixed values of γ and δ, $\frac{D}{\sigma^2}$ can be increased by increasing the feedback inhibition W_{EI}.

However, the reality is not so simple. In fact, since all the coefficients a,b,c,γ and δ depend on the fixed point

activities (E_0, I_0), changing the feedback inhibition W_{EI} will also change those coefficients. Then, increasing W_{EI} does not only increases the non-normal amplification, but also decreases the normal amplification by increasing the value of λ_0. This can be visualize in Fig.**??-Top** by the fact that if we vertically increase the y-axis of a point by keeping constant it x-axis, its distance to the Hopf bifurcation will increase, meaning that the corresponding value of λ_0 will increase. So, increasing W_{EI} will increase the non-normal amplification at the cost of decreasing the normal amplification and may lead to a resulting lower total amplification. Therefore, increasing the non-normal amplification, should be done at fixed value of λ_0. In other words when increasing W_{EI} we should make sure that λ_0 remains constant. This means that we should also increases W_{EE}. A realization of the non-normal amplification is achieved on the ascending direction of the curves $\lambda_0 = \lambda_i$. If we stay on those curves and increase both the recurrent excitation and the feedback inhibition, the network frequency will decrease, the firing rates and the total synaptic currents will also decrease. However, the parameters $\frac{D}{\sigma^2}$ increase. This is illustrated in Fig.**??-Bottom-(f-j)** for the four curves $\lambda_0 = \lambda_0^i$ plotted from the data. This allows to see that increasing values of the stimulus contrast leads to increasing values of the frequency, the population firing rates and the total synaptic currents. In contrast it leads to decreasing values of the non-normal amplification. Interestingly, we found that the stimuli with the largest Fisher Information have a total synaptic current close or equal to zero. A signature of the balance between the excitation and the inhibition. When increasing W_{EE} (the normal amplification) we can prevent the network to run into the linear instability by increasing the feedback inhibition W_{EI} such that λ_0 remains fixed. Clearly, increasing W_{EE} increases the level of excitation and bring the network towards the instability while increasing W_{EI} increases the level of inhibition and prevent the network to run into the instability. When strong recurrent excitation W_{EE} is stabilized by strong feedback inhibition W_{EI}, the result is a strong normal and non-normal amplification. This is also known as balanced amplification [23, 24] because the strong recurrent excitation is balanced by a strong feedback inhibition to produce such amplification. The network operating regime is known as Inhibition-Stabilized (IS) regime [25, 26]. The inhibition-stabilized regime is identified as the area between the vertical dashed line and the Hopf bifurcation. To compute the mean frequency we have used the following relation.

$$\omega_0 = \frac{1}{2}\sqrt{-(A_{EE} - A_{II})^2 - 4A_{EI}A_{IE}} \tag{12}$$

where the coefficients A_{EE}, A_{EI}, A_{IE} and A_{II} depends on the fixed points activities and the time-constant should be considered to identify the exact value of the amplification.

Mechanisms of stimulus-induced gamma rhythms

When the visual stimulus is presented to the monkey, it induces gamma rhythms in its primary visual cortex (or an area within it that we assume to be an E-I neural network). The properties (frequency, power,) of the induced gamma rhythms depend on the visual stimulus (contrast). Moreover, the stimulus onset shows a sharp transient of high power and large frequency variability (detected in the spectrogram after the average over several trials of the same stimulus presentation). Our analysis here has considered the data in the stationary state after the stimulus onset. The dynamical regimes inferred in the previous sections referred to the stationary states of the primary visual cortex after the stimulus onset. In other words, our analysis allows to understand the states of the primary visual

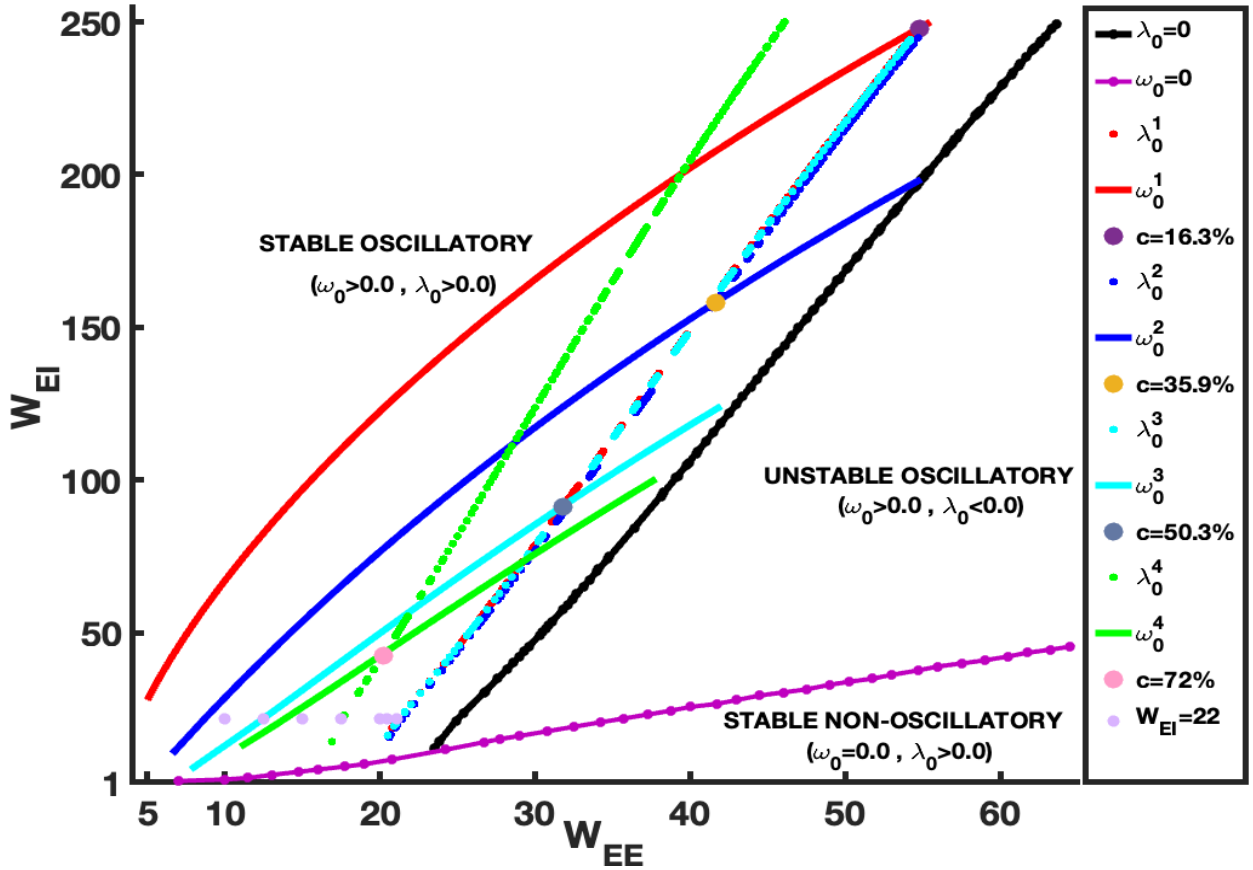

FIG. 5. **Stability diagram: Low-to-moderate stimuli cause transition to the Inhibition-Stabilized regime.** Bifurcation diagram in the recurrent excitation W_{EE} and feedback inhibition W_{EI} subspace, which is the natural subspace in which to investigate ISN [27] and which contrasts with our previous study of burst determinants [9]. The black dotted line (Hopf bifurcation) shows the boundary between the stable oscillatory and the unstable oscillatory regimes. It is the numerical solution of the equation $\lambda_0 = 0$. Similarly, the magenta dotted line shows the boundary between the stable non-oscillatory regime ($\lambda_0 < 0$ and $\omega_0 = 0$) and the oscillatory regimes (both stable and unstable). This line is obtained by solving numerically the equation $\omega_0 = 0.0$. The region between the vertical black dashed line and the Hopf bifurcation represents the IS regime. In this regime, strong feedback inhibition is required to prevent the network running into the instability. Note that here, the IS regime belong to the broad stable oscillatory regime. We focused our analysis on this stable oscillatory regime since it is the one that exhibits the variability seen in the data (see Methods). Solid colored lines represent the points where the network frequency ω_0 coincides with the frequency found in the data. Similarly, dashed colored lines match with the points where the network λ_0 meets the data. Colors red, blue, magenta and green refer to the parameters from stimuli 1 to 4 respectively. The intersection between the solid and dashed lines of the same color are marked in bold and correspond to the responses seen in the data. We have inserted 7 points at fixed value of the feedback inhibition W_{EI} but increasing value of the recurrent excitation W_{EE} to mimic the transition to the instability, therefore showing to the normal amplification. Points in the colored dashed lines have all the same value of λ_0 but increasing values of the feedback inhibition W_{EI} and those of $\frac{D}{\sigma^2}$. Then the dashed lines highlight the non-normal amplification. These two amplifications are more elaborated in the next figure.

cortex after the stimulus onset, but it does not explicitelly explain the history of the primary visual cortex before and during the stimulus onset (since we haven't studied the data during those epochs). Therefore, the mechanism of the stimulus-induced gamma rhythms in the monkey's primary visual cortex is still unclear. To elucidate it, we infer the values of the external inputs in our E-I model for different values of the stimulus contrast (see Fig.6**A**). We also identified the different dynamical regimes and confirmed that the stimulated states were in the stable oscillatory regime previously identified. To better understand the role of the stimulus contrast, we include two additional datasets obtained for the values of the contrast at 6.1 and 9.7 percents. These datasets were not previously considered because their mean frequencies are outside of the gamma band used in our analysis. From the locations of the points

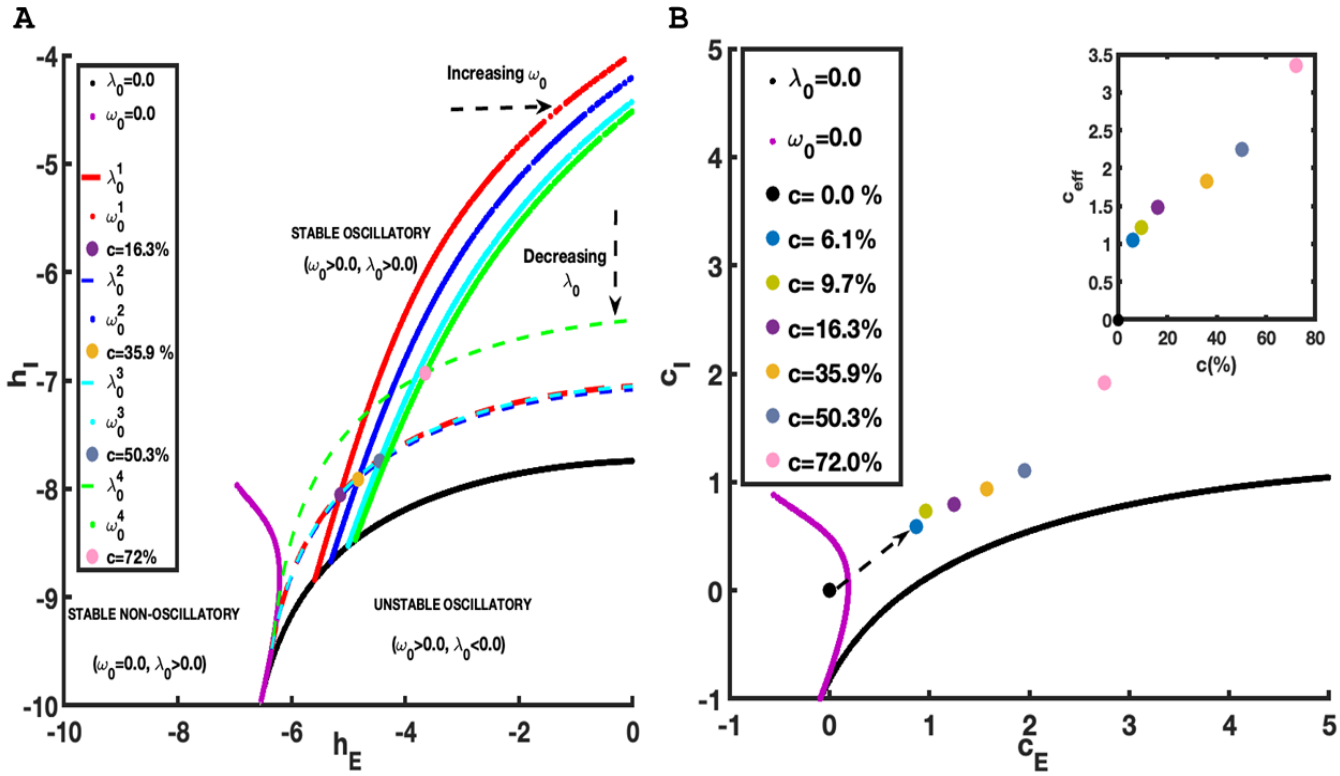

FIG. 6. **Stimulus contrast inference and Mechanisms of stimulus-induced gamma rhythms in the monkey's V1 area. A**: External inputs inference. The dotted and dashed curves are those where ω_0 and λ_0 are respectively constants and equals to the values extracted in the four datasets. The colors red, blue, cyan and green refer to the responses induced by the stimuli one to four respectively. The dotted black curve is obtained when $\lambda_0 = 0$, and separates two distinct dynamical regimes; an unstable oscillatory regime where oscillations are linearly unstable and a stable oscillatory regime where oscillations are linearly stable. The dark red curve corresponds to $\omega_0 = 0$ and separates a stable non-oscillatory regime (also known as the asynchronous state) where the model does not oscillate, from oscillatory states (both linearly stable and unstable). The external inputs corresponding to the responses (data) induced by the stimuli one to four are the intersection points of the curves (dashed and dotted) in red, blue, cyan and green respectively. **B**: Mechanisms of stimulus-induced gamma rhythms. We extended the approach in **A** to two additional responses induced by stimuli at contrasts 6.1 and 9.7 percent (see dots in blue and dark green respectively). These responses (data) were not included in the previous analyses because their corresponding mean frequencies were not in the gamma band chose here. Considering the fact that the external inputs to the E and I populations are the sums of baseline inputs coming from others neural networks and the stimuli, we centered our frame at the baseline inputs represented by the black dot. The contributions of the stimulus contrast to the E and I populations are modelled by c_E and c_I respectively. In the new frame, we can now visualize that a stimulus with higher contrast leads to higher values of c_E and c_I, and a higher value of the frequency (see the direction of increasing ω_0 in panel **A**). Interestingly, it can be seen that the baseline inputs correspond to a dynamics in the stable non-oscillatory or asynchronous state. The stimulus therefore induces a transition from the stable non-oscillatory to the stable oscillatory states. The higher the value of the contrast, the higher the value of the frequency of the response. A result that agrees with in-vivo recordings. If we further define the effective contrast c_{eff} as the Euclidian distance of each response in the new frame, then we see that the effective contrast increases as the stimulus contrast does (see inset figure).

corresponding to the six datasets, we can infer the pre-stimulus condition (corresponding to the unstimulated state of the monkey) position in the stable non-oscillatory state but close to the transition to stable oscillatory state. We therefore centered our frame to the reference state (see Fig.6**B**). In the new frame, we can visualize the trajectory of the response for different values of the stimulus contrast (see the dashed arrow). Gamma rhythm responses induced by low values of the stimulus contrast are closer to the stable non-oscillatory regime and those with larger values of the stimulus contrast are farther away from it. In addition, the network frequency increases when the dynamics move away from the stable non-oscillatory regime (see the direction of increasing ω_0 values in Fig.6**A**). This is consistent with

recordings since gamma rhythm responses induced by higher stimulus contrasts will have higher mean frequencies. Therefore, we propose that before the stimulus onset (pre-stimulus period), the V1 area is in a stable non-oscillatory regime (consistent with the asynchronous state reported in the literature) (there is no gamma rhythms). When the stimulus is presented, depending on the value of its contrast, it induces gamma rhythms through a transition from the stable non-oscillatory to the stable oscillatory regimes. The sharp transient at the stimulus onset(detected in the trial average response spectrogram) is the signature of the transition between the two stable states. The stimulus triggers an oscillatory response which can be optimally encoded or not. The optimal encoding capacity is achieved when the stimulus induces a response in the Inhibition-Stabilized regime. Precisely, the stimulus acts differently on the excitatory and inhibitory populations. We have denoted by c_E and c_I the contributions of a specific stimulus contrast to the E and I populations respectively. The effective value of the contrast on both the E and I populations can be quantified as the Euclidian distance of each responses in the new frame (see inset Fig.6**B**). For weak values of the stimulus contrasts, the response is amplified by the excitation, for moderate values, the inhibition stabilizes the excitation and the response are driven by strong fluctuations resulting to high information encoding and for large values of the stimulus contrasts the inhibition dominates the network and the corresponding response is dampened (response to the stimulus at 72 percent contrast). Our results suggest that a stimulus can induce a change of state in the cortex.

Stimulus dependency and modulation of gamma frequency. The most prominent effects characterizing the mechanisms of stimulus-induced gamma rhythms are the sharp transient at stimulus onset [1, 28], as well as the dependency and modulation [1] of the gamma frequency by the stimulus contrast. Here, we have argued that the sharp transient at the stimulus onset is the signature of the transition from the stable non-oscillatory to the stable oscillatory regimes. We have also shown that increasing the contrast in our model (precisely the effective value c_{eff} or both c_E and c_I) also increases the rhythm frequency. To further validate our predictions, we have performed simulation on our E-I neural network using a time-dependent stimulus. We chose different strengths and profiles of the stimulus (constant, monotonically increasing and double Gaussian). The results show that increasing the stimulus from zero to a positive value allows to induce a state transition. In addition, the state transition captures well the sharp transient observed in several recordings [1, 28]. The resulting gamma rhythm frequency after the stimulus onset is modulated according to the strength and the profile of the stimulus. Specifically, a stimulus with a constant contrast will lead to a constant gamma frequency, one with a linearly increasing contrast to linearly increasing frequency, and a stimulus with a double Gaussian contrast profile will lead to a double Gaussian-shaped frequency. Thus, our hypothesis about the mechanism of stimulus-induced gamma rhythm qualitatively agrees with previous observation and recordings.

II. DISCUSSION

*Our analysis and interpretation of the roles of λ_0 and D are limited by the number of stimuli used to generate the gamma rhythms.

*Cite Ainsworth-Whittington Neuron review 2012: oscillatory encoding of strength into rhythm (temporal coding)

*Burst properties mirror Fisher info results

*OUTLOOK: Muller, L., Reynaud, A., Chavane, F., and Destexhe, A. (2014). The stimulus- evoked population response in visual cortex of awake monkey is a propagating wave. Nat. Commun. 5, 3675. https://doi.org/10.1038/ncomms4675.

The high variability of brain activity has made the identification of brain states and transitions between states challenging. Our analysis reveals that state transitions can be induced in primary visual cortex by static gratings, and in particular, gamma rhythms arise when the contrast is not too strong. Our fitting procedure is based on a parsimonious envelope-phase description of the amplitude and phase of the gamma rhythm, extracted from a standard rate model description of effective E-I PING dynamics in the cortical circuits contributing to the recorded LFPs. The benefit of this model is the massive reduction in number of parameters from the original E-I system to two meta-parameters, namely, the stability parameter λ_0 and the effective noise intensity D. These can be estimated from the data, along with the mean frequency ω_0 of the fast underlying rhythm, using derived properties of the model and maximum likelihood estimation.

According to our results, there is no gamma rhythm prior to stimulus onset, with the network being in a stable asynchronous non-oscillatory state [29]. The stimulus induces a transition from this state to a stable asynchronous oscillatory state. As in [3, 9], the rhythm caused by the stimulus reflects a stable focus-type attractor of the E-I dynamics, making it induced (i.e. reliant upon for its existence) rather than perturbed by neuronal noise (Fig.3**E-F**). In this picture, the sharp transient of large amplitude and frequency variability arises from the transition between the asynchronous non-oscillatory and oscillatory states. Our proposed mechanism produces the stimulus-dependent modulation of gamma rhythm frequency reported in previous studies [1, 28]. The fitted envelope-phase model also exhibits similar burst-feature variability (duration, amplitude and frequency) to that of the real data across stimuli.

We quantified the encoding capacity of contrast into gamma oscillations via the Fisher Information curve. Our theory shows that the FI is proportional to the square of the second moment of the amplitude fluctuations. A maximal value of the FI occurred for a moderate contrast value, reached moderate values for neighhboring neighboring contrasts. Perhaps surprisingly, the smallest FI value occurred for the strongest contrast (Fig.1**C**), meaning that the decoding of the grating contrast feature had minimal information at large contrast. Furthermore, we could link the FI magnitude to the burst-features and the fitted parameters. For instance, FI correlated positively with mean burst duration. This duration could then be used as a biomarker of stimulus encoding capacity.

The inferred envelope-phase model parameters offers insight into the mechanism of gamma rhythm amplification, which we have found actually involves two mechanisms. The first is the normal amplification and is related to the stability parameter λ_0 approaching zero. The second one is a non-normal amplification that happens when the noise D parameter increases. These two parameters and associated amplification mechanisms explain well the total stimulus-induced amplification found in the data. Counter-intuitively, a monotonically increasing stimulus strength was found to change implicate both kinds of amplification; this is accompanied by a simple increase in mean gamma frequency. It remains to be seen if this rule applies to other stimulus determinants of gamma power. The non-normal amplification found here is commonly found in cortex [23, 25, 30]. Here we were able to tie gamma burst properties with the level of non-normal amplification in the data and the corresponding E-I network. Thus, our results agree well with previous ones pointing to ISN as an optimal operating regime of the cortex.

Using the envelope-phase model across stimuli, we could also indirectly estimate two synaptic coefficients in the

EI network that impact the ISN status, namely, the recurrent excitation W_{EE} and feedback inhibition W_{EI}. This enhances our biophysical interpretation the two amplification mechanisms uncovered using the envelope-phase model. The normal amplification is related to the increasing recurrent excitation that leads to heightened network instability. On the other hand, the non-normal amplification is observed for both the recurrent excitation and the feedback inhibition increasing on a specific trajectory, and for a specific noise intensity. Furthermore, we have identified a dynamical regime known as the Inhibition stabilized (ISN) regime characterized by a strong recurrent excitation stabilized by a strong feedback inhibition, resulting in an overall high network amplification. The three stimulated states corresponding to moderate-to-high values of the FI were in the ISN regime, whereas the stimulated state with the minimal FI fell outside this regime. Clearly, optimal information encoding capacity is associated with the ability for a stimulus to trigger a response into the ISN regime with a (near-zero) balance between synaptic excitation and inhibition. In addition, network analysis shows that the normal and non-normal amplification are distinct mechanisms that entail opposite variations of the network frequency and the mean E-I population rates with stimulus strength.

Results with other data will deviate somewhat from those in our study, e.g. the FI may look qualitatively similar but differ quantitatively. Some data may also show a monotonic increase in gamma power [1], or the stimulus-induced dynamics may straddle the Hopf bifurcation, enabling noisy limit cycle activity to encode contrast (refer to Ray's recent PLOSCB?). In this latter case, a similar (but more involved) analysis based on our theoretical extension of the envelope-phase model to the limit cycle side could be attempted [22]. Nevertheless, the fact that the encoding capability of the stimuli can be quantified by a simple envelope-phase model derived from a noise-driven linearized E-I circuitry is quite surprising. Equally surprising is the fact that only three parameters were necessary to explain the LFP and burst variability present in the data. Previous work has shown that such as model can explain observed variability, such as that of gamma burst features in LFPs of awake or anesthetized monkeys [3]. Similarly, a noise-driven harmonic oscillator could explain the cycle-to-cycle variability observed in monkeys' LFPs [17]. However, we are not aware of any study that used a stochastic envelope-phase model, namely one with a highly reduced parameter space, to establish a relationship between stimulus and response.

Our results suggest that working memory and interareal communication may also been linked to the ability of the brain to optimally encode external inputs. In this picture, gamma bursts may be seen as epochs of time during which information is retained in a working memory task [14], or transmitted/received to/by another brain region during communication. In this picture, longer burst durations enhance the task. However, increasing the stimulus contrast does not always increase the FI nor the mean burst duration. This raises the question of how to choose, in e.g. a therapeutic context, a stimulus strength that triggers or modulates gamma power with desired mean burst duration. Our study provides insight into achieving this goal.

Another circuitry that supports gamma rhythms is the Interneuron Network Gamma (ING) which relies on the interaction of inhibitory neurons [31]. It can generate rhythms with similar variability than in vivo data. Whether the specific data analyzed here are generated by an ING or a PING network is hard to discriminate. However, the PING mechanisms is analytically more tractable and suited for data analysis. In addition, it allows one to formulate synaptic mechanisms behind gamma oscillation amplification. Also, we have shown that the optimal stimulated states corresponded to the E-I networks operating in the Inhibition-Stabilized regime. This suggests an interaction between excitation and inhibition, and points towards the PING mechanism. The IS regime with the balance between excitation and inhibition has been identified in a number of theoretical and experimental studies as the optimal operating regime

of the cortex [25–27, 32]. Both types of studies have shown that gamma rhythms are likely generated in the ISN regime [33, 34]. Such studies assumed E-I deterministic limit cycle attractor dynamics. In contrast, our study connects the ISN regime in a noisy E-I network to a stable focus attractor and appends an information-theoretic analysis of encoding into Gamma envelope features.

Our work provides a plausible mechanism by which a stimulus induces gamma rhythms response in the monkey's primary visual cortex. Before, the stimulus onset, there are no gamma rhythms. The network is likely to be in the asynchronous state [29] characterized by a stable fixed-point attractor. When a stimulus with enough contrast is on, it modifies the state of the V1 area network that generates gamma rhythms by inducing a bifurcation from the asynchronous state with no rhythms to another asynchronous state with gamma rhythms. The bifurcation between the two states is characterized by a sharp transient of large frequency and amplitude variability seen in the trial-average spectrogram. The amplification of the stimulated state (stationary state after the transient stimulus onset) depends on the value of the contrast. In fact, increasing the stimulus contrast linearly increases the frequency of the gamma rhythms, the mean population firing rate of the E and I cells, and the synaptic coefficients (Precisely, the effective recurrent excitation and feedback inhibition) of the network. We have supposed that the noise in the V1 area or its corresponding E-I network comes from random synaptic inputs of neurons in the same and other similar nearby networks. It is reasonable to choose the intensities of these noisy inputs proportional to the mean population firing rates, then, such intensities also increase as the stimulus contrast does. The non-normal amplification parameter in our study decreases as the gamma frequency increases but increases as the feedback inhibition and the noise intensity of the network do. When the stimulus is triggered, it induces several network mechanisms at the same time. Some of them have similar effects as the level of noise (or mean population firing rates) and the balance between excitation and inhibition, that increase the non-normal amplification. Other mechanisms like the increasing of the network frequency have the opposite effect. The summary of these effects on the network suggests that gamma rhythms amplification is a complex phenomenon that involves several primary mechanisms. Each stimulated state corresponds to the coordination of all these primary mechanisms. The dominance or weakness of one or several of these primary mechanisms could lead to pathological gamma rhythms. For instance, the imbalance between the excitation and the inhibition leads to a pathological state outside of the IS regime. At the other side, an adequate coordination of all these mechanisms lead to optimal states in the IS regime. However, even in the IS regime, some states can be more amplified than other depending on the contribution of all these primary mechanisms.

To mimic the effect of the stimulus contrast, we added a constant drive to the population of excitatory neurons only. The value of the external drive is the representation of the contrast and allows to reproduce the behaviors seen in the data and the mechanisms of stimulus-induced gamma rhythms. When the external drive becomes time-dependent, it modifies the frequency of the gamma rhythms accordingly similar to previous recordings [1, 28]. The mechanisms of stimulus-induced gamma rhythms described here is also consistent with the limit cycle regime recently found in the data [34]. Indeed, according to our modelling, a constant drive applied to the excitatory neurons can also induce a transition from the stable asynchronous state to the limit-cycle state (called the unstable oscillatory regime here, since our analysis is linear). Such a transition also induces a sharp transient of large amplitude and frequency variability at the stimulus onset and higher harmonics in the LFP power spectra of the stimulated state like in [34]. When the attractor is a limit-cycle, the Inhibition-Stabilized regime has to be interpreted by taking into account nonlinear effects and the superlinear effect of the inhibition [33, 34]. The modulation of gamma rhythms frequency by the stimulus

contrast can also be achieved when the attractor of the network is a limit-cycle. Therefore, the attractor of the V1 area depends on the stimulus. This may explain why some data can suggest a stable fixed point attractor [3, 17] while other a limit-cycle attractor [34]. The attractor of gamma rhythms in the V1 area is stimulus-driven. This shows the importance of optogenetic methods for the identification of the dynamical regime of the cortex. Future studies should extend the computation of the Fisher Information done here for gamma rhythms with a limit-cycle attractor. This could be done by using a recent envelope-phase model of noise-perturbed oscillations [22]. Other avenues will be to extend the analysis to other stimulus properties like the orientation of the contrast or more complex stimuli including moving and colored stimuli.

The transient observed during the stimulus onset will be consider during a future study.

III. METHODS

Model. The monkey's primary visual cortex contains connected excitatory and inhibitory cells which generate gamma oscillations at the population level. We are not interested in the spiking activity of individual cells, but rather the population firing rate, first in the steady state, and second, at stimulus onset [1]. We seek a neural network that can generate oscillations with properties similar to those seen experimentally. We consider a stochastic version [19] of the Wilson-Cowan model [18] for oscillation generation. The solution of this two-dimensional model with additive noises describes the fraction of excitatory (E) and of inhibitory (I) cells that are active as a function of time. The mean synaptic coupling from E cells to themselves, and from E cells to I cells, are given respectively by W_{EE} and W_{IE}. Similarly, the mean synaptic coupling from I cells to themselves and from I cells to E cells are W_{II} and W_{EI}. The E cell population receives a constant external stimulus h_E (that will be clarified below), while the I cell population is driven by the constant external current h_I. Additive noise is assumed to arise mainly from the randomness of synaptic events in the network. A schematic of the network is shown in Fig.7**A**. The stochastic WC model (SWC) is:

$$\frac{dE(t)}{dt} = -\alpha_E E(t) + \big(1 - E(t)\big)\beta_E f(s_E(t)) + \sigma_E \xi_E(t) \tag{13}$$

$$\frac{dI(t)}{dt} = -\alpha_I I(t) + \big(1 - I(t)\big)\beta_I f(s_I(t)) + \sigma_I \xi_I(t)\,. \tag{14}$$

The sigmoid function is $f(x) = (1 + e^{-x})^{-1}$. The total excitatory $s_E(t)$ and inhibitory $s_I(t)$ synaptic inputs to a neuron are given respectively by

$$s_E(t) = W_{EE}E(t) - W_{EI}I(t) + h_E \tag{15}$$

$$s_I(t) = W_{IE}E(t) - W_{II}I(t) + h_I\,, \tag{16}$$

where $\xi_{E,I}$ are two independent Gaussian white noises with intensities σ_E and σ_I and the following properties:

$$< \xi_{E,I}(t) > = 0, \quad < \xi_E(t)\xi_I(t) >= 0$$
$$< \xi_E(t^{'})\xi_E(t) > =< \xi_I(t^{'})\xi_I(t) >= \delta(t - t^{'})\,.$$

The external input to the excitatory and inhibitory population can be written respectively as

$$h_E = h_E^0 + c_E \tag{17}$$

$$h_I = h_I^0 + c_I \,. \tag{18}$$

In this formulation h_E^0 and h_I^0 are the baseline inputs (i.e. prior to the onset of the stimulus contrast) to the E and I populations. The parameters c_E and c_I represent the contributions of the stimulus. The noise-free ($\sigma_E = \sigma_I = 0$) dynamics has a fixed point (E_0, I_0) which, for the parameter values chosen in our work, can be a stable or an unstable focus. If the fixed point is a stable focus, deviations from the fixed point converge to zero. The noise slowly erases the memory of the initial conditions, and induces deviations from the fixed point that relax in a noisy oscillatory manner [7, 9]. Adding noise to this regime leads to noise-induced oscillations or quasi-cycles. If the fixed point is an unstable focus, the system undergoes a Hopf bifurcation to a stable limit cycle. The noise then yields a stochastic limit cycle [35, 36] by perturbing the amplitude, frequency and phase of the deterministic oscillation.

Dynamical regime of Gamma oscillations found in monkey area V1. The SWC exhibits two distinct types of gamma oscillations separated by the Hopf bifurcation (black doted line in Fig.7**B**). Below the bifurcation, oscillations are induced by the noise (Fig.7**B-Left**). Above the bifurcation, oscillations exist even in the deterministic limit, a weak noise acts as a perturbation we call this regime noise-perturbed (Fig.7**B-Right**). We are interested in the dynamical regime that show similar variability that gamma rhythms found in the V1 primary visual cortex. By looking at the spectrograms and time-series (Fig.7**C**) of rhythms generated in the two regimes and from previous analyses [9], we concluded that the noise-induced regime is more likely to be the operating regime of monkey V1 responding to static visual stimuli. We therefore focus our analysis in the noise-induced regime. Our hypotheses are as follows:

- Before the stimulus (this corresponds to c_E=c_I=0 in Eqs.17)-18 the network does not show oscillations. The presence of a sufficient stimulus induces a transition from a state without oscillations (Stable non-oscillatory) to the noise-induced oscillations (Stable oscillatory) state. However we consider that such transition has already happened for the four stimulus contrasts studied (except in Fig.6) and we focus on the post-stimulus epoch.

- We do not specifically model the form of stimulus, but we assume that its contrast is contained in the external current $h_{E,I}$ as defined in Eq.17-18.

- At this stage, we are not trying to map the stochastic wilson cowan with the monkey's primary visual cortex data directly. Instead we are looking at the broad dynamical regime that show similar variability.

- Since noise is critical in the noise-induced regime, we assumed that information about the stimulus is contained in the deviations from the fixed points or mean activities. We consider such deviations as measures of local field potentials.

From numerical simulations of excitatory and inhibitory fluctuations or Local field potentials, we see that short epochs of synchrony or gamma bursts match with epochs of high envelope values (Fig.7**C-Left**). This suggests that the envelope of gamma rhythms fluctuations could be appropriate for the analysis of bursts. Before, an analysis of

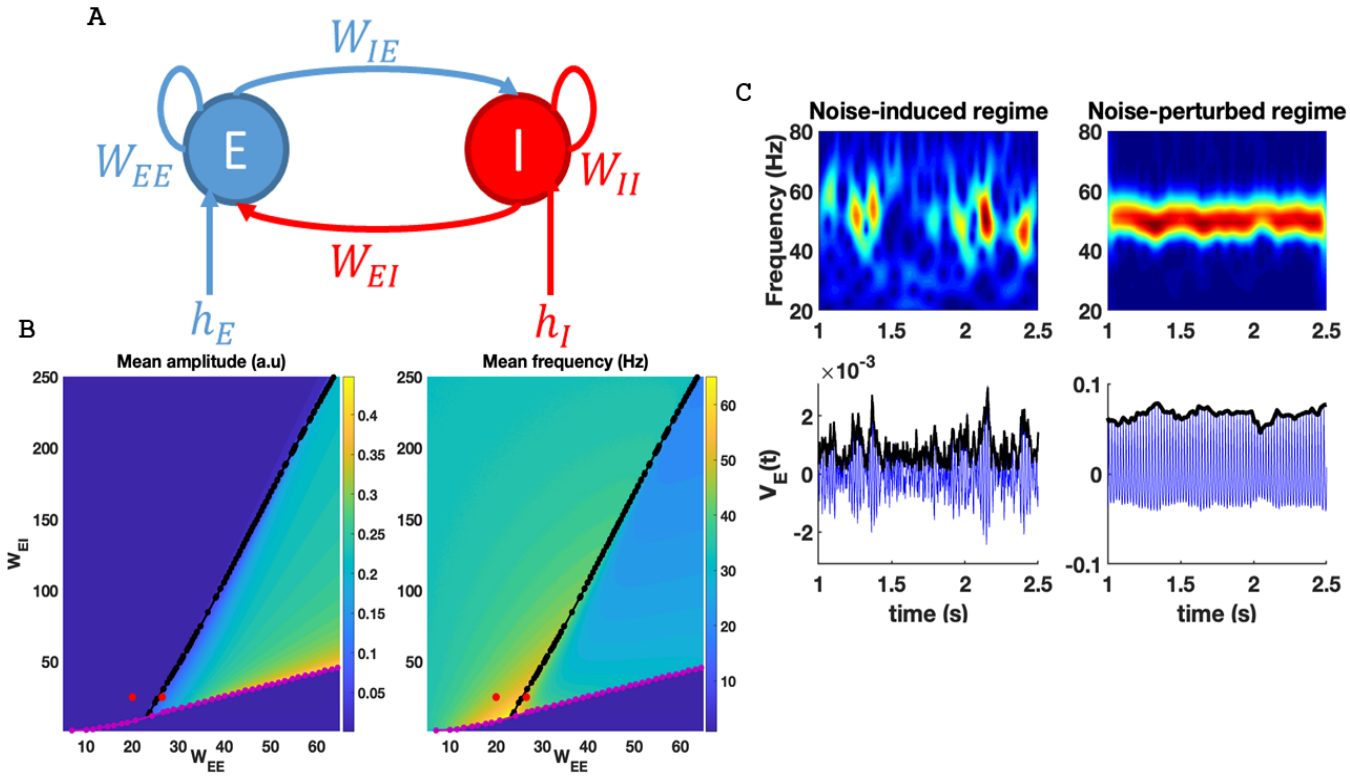

FIG. 7. **Gamma oscillations in the Stochastic Wilson-Cowan model. (B)**. **(Left)**: Synchronization level of the E cells in the subspace of recurrent excitation W_{EE} and feedback inhibition W_{EI} parameters. The synchronization is measured from numerical simulations of the activities Eqs.26 as the average of the difference between each peak and its successive trough. Specifically, we used a Gaussian filter and the Matlab function "findpeaks" to filter and extract the peaks and troughs of the LFPs. The curve in black represents the Hopf bifurcation boundary computed by setting the real part $-\lambda_0$ of the complex conjugate eigenvalues to zero, and the curve in magenta shows the transition between $\omega_0 = 0$ and $\omega_0 > 0$. The two magenta dots lie on either side of the Hopf boundary at $W_{EI} = 25$ and $W_{EE} = 20$ (noise-induced) or $W_{EE} = 26.5$ (noise-perturbed). **(Right)**: Mean oscillation frequency of the oscillations generated in the Left panel. **(C)-Left**: Time-series (**Top**) and Spectrogram **(Bottom)** corresponding to the red dot in the noise-induced regime in **(B)**. Oscillations appear as discrete (short) epochs of synchrony called "bursts"; the frequency content also fluctuates from burst to burst. **(Right)**: Timeseries (**(Top)**) and Spectrogram (**Bottom**) for the red dot in the noise-perturbed regime in panel **(B)**. Oscillations are now highly coherent with peak power around $f = 50Hz$. Parameters are $W_{II} = 1.3, W_{IE} = 32, h_E = -3.8, h_I = -8, \alpha_E = 0.1, \alpha_I = 0.13, \beta_E = 1, \beta_I = 2$, $\sigma_E = 0.002$ and $\sigma_I = 0.002$)

the properties of the network without noise and with noise is required.

Dynamics of the fluctuating local field potentials. We wish to characterize the fast and slow features of the deviations from the fixed point in the noise-induced regime. We look for solutions of Eqs.(13-14) in the form:

$$E(t) = E_0 + V_E(t) \qquad\qquad I(t) = I_0 + V_I(t) \tag{19}$$

where the fixed points E_0 and I_0 solve the deterministic equations:

$$-\alpha_E E_0 + \big(1 - E_0\big)\beta_E f(s_{E_0}) = 0 \tag{20}$$

$$-\alpha_I I_0 + \big(1 - I_0\big)\beta_I f(s_{I_0}) = 0\,, \tag{21}$$

with the definitions

$$s_{E_0} = W_{EE}E_0 - W_{EI}I_0 + h_E$$
$$s_{I_0} = W_{IE}E_0 - W_{II}I_0 + h_I\,.$$

Note that we have restricted our analysis to the parameter regime most relevant to brain rhythms, namely the one where the system has only one deterministic fixed point. We inserted Eqs. (19) into Eqs. (13-14) and used the following Taylor expansion of the sigmoid function:

$$f(s_E(t)) = f(s_{E_0} + \delta s_E(t)) \tag{22}$$
$$= f(s_{E_0}) + \delta s_E(t) f^{'}(s_{E_0})$$
$$f(s_I(t)) = f(s_{I_0} + \delta s_I(t)) \tag{23}$$
$$= f(s_{I_0}) + \delta s_I(t) f^{'}(s_{I_0})$$

where

$$\delta s_E(t) = W_{EE}V_E(t) - W_{EI}V_I(t) \tag{24}$$
$$\delta s_I(t) = W_{IE}V_E(t) - W_{II}V_I(t). \tag{25}$$

The Taylor expansion made above in Eqs. (22-23) is accurate for small deviations in Eqs. (24-25). The fluctuations $V_E(t)$ and $V_I(t)$ are considered as local field potentials (LFP) in the context of brain dynamics [7, 9]. These LFPs are dimensionless in the Wilson-Cowan modeling framework since they are simply deviations from the steady state fractions E_0 and I_0; they are nevertheless interpreted as potentials [3]. We then replace $f(s_E)$ and $f(s_I)$ by their Taylor expansion at the first order given in Eqs.22-23. Taking into account the fixed point solutions Eqs.20-21 and keeping only linear terms, the equations above reduce to:

$$\frac{dV_E(t)}{dt} = A_{EE}V_E(t) + A_{EI}V_I(t) + \sigma_E\xi_E(t) \tag{26}$$
$$\frac{dV_I(t)}{dt} = A_{IE}V_E(t) + A_{II}V_I(t) + \sigma_I\xi_I(t)\,, \tag{27}$$

with the following parameter definitions:

$$A_{EE} = -\big[\alpha_E + \beta_E f(s_{E_0}) - (1-E_0)\beta_E f^{'}(s_{E_0})W_{EE}\big]$$
$$A_{II} = -\big[\alpha_I + \beta_I f(s_{I_0}) + (1-I_0)\beta_I f^{'}(s_{I_0})W_{II}\big]$$
$$A_{EI} = -(1-E_0)\beta_E f^{'}(s_{E_0})W_{EI}$$
$$A_{IE} = (1-I_0)\beta_I f^{'}(s_{I_0})W_{IE}\,.$$

The system of Eqs. (26-27) can be rewritten as

$$\tau_E \frac{dV_E(t)}{dt} = -V_E(t) + W_{eef} V_E(t) - W_{eif} V_I(t) + \sigma_{E_f} \xi_E(t) \tag{28}$$

$$\tau_I \frac{dV_I(t)}{dt} = -V_I(t) + W_{ief} V_E(t) - W_{iif} V_I(t) + \sigma_{I_f} \xi_I(t)\,, \tag{29}$$

where the associated "effective" coefficients are given as

$$W_{eef} = E_0\big(1-E_0\big)\left[1 - \frac{\alpha_E E_0}{(1-E_0)\beta_E}\right] W_{EE} = g(E_0) W_{EE} \tag{30}$$

$$W_{iif} = I_0\big(1-I_0\big)\left[1 - \frac{\alpha_I I_0}{(1-I_0)\beta_I}\right] W_{II} = g(I_0) W_{II} \tag{31}$$

$$W_{eif} = E_0\big(1-E_0\big)\left[1 - \frac{\alpha_E E_0}{(1-E_0)\beta_E}\right] W_{EI} = g(E_0) W_{EI} \tag{32}$$

$$W_{ief} = I_0\big(1-I_0\big)\left[1 - \frac{\alpha_I I_0}{(1-I_0)\beta_I}\right] W_{IE} = g(I_0) W_{IE} \tag{33}$$

$$\tau_E = \frac{1-E_0}{\alpha_E}, \qquad \tau_I = \frac{1-I_0}{\alpha_I} \tag{34}$$

$$\sigma_{E_f} = \frac{1-E_0}{\alpha_E}\sigma_E, \quad \sigma_{I_f} = \frac{1-I_0}{\alpha_I}\sigma_I\,. \tag{35}$$

$$\tag{36}$$

The functions $g(x) = x\big(1-x\big)\big[1 - \frac{\alpha_x x}{(1-x)\beta_x}\big]$ can be interpreted as gain functions. The precise values of these functions depend on the fixed points E_0 and I_0 which themselves depend on the external stimulus applied to the network. Basically, the external input modulates the original synaptic coefficients, leading to effective synaptic coefficients which are now stimulus-dependent. For our analytic work, we will use Eqs. (26-27) instead of Eqs. (13-14). Equations (26-27) are the linearized approximation of the stochastic Wilson-Cowan model, denoted henceforth as the linear SWC. In the noise-induced regime the linear SWC is a good approximation of the full nonlinear SWC [9]. Moreover, recent studies have shown that the linear SWC could be used to generate gamma rhythms similar to those found in the monkey V1 area [3, 17]. Also, gamma rhythms in-*vivo* show general properties as a spiking delay between excitatory and inhibitory neurons. We focus on the linear SWC dynamics. We are interested in the slow envelope dynamics of the excitatory and inhibitory cells.

Mean Firing rates. The mean firing rates of the E and I populations can be respectively defined as [19]:

$$r_E(t) = \Big(1 - E(t)\Big)\beta_E f(s_E) \tag{37}$$

$$r_I(t) = \Big(1 - I(t)\Big)\beta_I f(s_I)\,. \tag{38}$$

Taking into account the fixed point conditions Eqs.20-21 and the stability of the noise-induced regime described in

the previous section, these equations can be further written as

$$r_E = \alpha_E E_0 \tag{39}$$

$$r_I = \alpha_I I_0 \tag{40}$$

The mean firing rates are proportional to the fixed points.

Noise variations. We have assumed that the noise strengths in the SWC represent the intensities of the stochastic firing of neurons in the network. The noise power spectra can therefore be chosen as proportional to the mean firing rates as follows.

$$\sigma_E^2 \propto \alpha_E E_0 \tag{41}$$

$$\sigma_I^2 \propto \alpha_I I_0 \tag{42}$$

This suggests that states with high firing rates correspond to those with high noise levels. This make sense since the noise corresponds to the stochastic firing of neurons. The choice of the noise power spectra proportional to the firing rates in the noise-induced regime has been proven in [19] by linear noise approximation to a slightly different SWC model with multiplicative noise. Here, we have assumed for simplicity the same noise strength for both the E and I population. That is

$$\sigma^2 = \sigma_E^2 = \sigma_I^2 \propto \alpha_E E_0 \propto \alpha_I I_0 \tag{43}$$

Hence, we expect the noise to be stronger as the firing rates increase.

Linear Analysis and properties of the Linear SWC dynamics. We assume that the noise as a perturbative effect on the intrinsic properties of the linear SWC. In other words, properties of the linear SWC are present even without the noises. Specifically, we are looking for properties between the amplitudes and phases of the E and I dynamics. For that, we perform the linear analysis of Eqs. (26-27). The associated noise-free linear system of Eqs. (26-27) can be written in the following matrix form:

$$\frac{d\mathbf{V^0(t)}}{dt} = \mathbf{A}\mathbf{V^0(t)}$$

where

$$\mathbf{V^0(t)} = \begin{bmatrix} V_E^0(t) \\ V_I^0(t) \end{bmatrix} \quad and \quad \mathbf{A} = \begin{bmatrix} A_{EE} & A_{EI} \\ A_{IE} & A_{II} \end{bmatrix} .$$

We look for a trial solution in the form:

$$\begin{bmatrix} V_E^0(t) \\ V_I^0(t) \end{bmatrix} = \begin{bmatrix} \tilde{B_E} \\ \tilde{B_I} \end{bmatrix} e^{\lambda t}$$

where $\tilde{B_E} = B_E e^{j\theta_E}$ and $\tilde{B_I} = B_I e^{j\theta_I}$. The eigenvalue λ of the associated matrix A is found by substituting the trial solution into the linear system, yielding

$$\frac{\tilde{B_I}}{\tilde{B_E}} = \frac{A_{EE} - \lambda}{-A_{EI}} = -\frac{A_{IE}}{A_{II} - \lambda} \quad .$$

The second equality leads to

$$\lambda = \frac{1}{2}(A_{EE} + A_{II}) \pm \frac{j}{2}\sqrt{-(A_{EE} - A_{II})^2 - 4A_{EI}A_{IE}} \quad .$$

We rewrite the eigenvalue in the compact form

$$\lambda = -\lambda_0 \pm j\omega_0$$

with stability parameter

$$\lambda_0 = -\frac{A_{EE} + A_{II}}{2} \tag{44}$$

and mean frequency of the LFP

$$\omega_0 = \frac{1}{2}\sqrt{-(A_{EE} - A_{II})^2 - 4A_{EI}A_{IE}}\,. \tag{45}$$

This leads to the exact expression of the amplitude ratio between the I and E fluctuations for the linearized SWC system:

$$\alpha = \left|\frac{\tilde{B_I}}{\tilde{B_E}}\right| = \frac{B_I}{B_E} = \sqrt{\frac{A_{IE}}{-A_{EI}}} \tag{46}$$

Similarly, the phase difference δ can be obtained from the relation

$$\delta = \theta_I - \theta_E = Arg\left(\frac{\tilde{B_I}}{\tilde{B_E}}\right). \tag{47}$$

For the parameters used here, we found

$$\delta = \theta_I - \theta_E = \arctan\left(\frac{-2\omega_0}{A_{EE} - A_{II}}\right). \tag{48}$$

In Eqs. 46-47, $||$ and Arg are respectively the modulus and the argument of the complex number $\frac{\tilde{B_I}}{\tilde{B_E}}$. In the presence of noise, one can compute amplitudes and phases of the E and I fluctuations from simulated time series using the analytic signal technique. The amplitude ratio and the phase difference are obtained by the following approximations:

$$\alpha = \frac{B_I}{B_E} \approx \left\langle \frac{Env\big[V_I(t)\big]}{Env\big[V_E(t)\big]} \right\rangle \quad and \quad \delta = \theta_I - \theta_E \approx \left\langle Arg\big[V_I(t)\big] - Arg\big[V_E(t)\big] \right\rangle.$$

Here $\langle . \rangle$ can be considered a time average of the stochastic process in Eqs. (26-27). Env is defined as the amplitude of the analytic signal associated with the LFP. For example, the analytic signal corresponding to $V_E(t)$ is

$V_E(t) + jH\big[V_E(t)\big]$, with the Hilbert transform H defined as

$$H[x] = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{x(\tau)}{t-\tau} d\tau \,,$$

where P signifies the Cauchy principal value. The amplitude of the stochastic signal is then $Env[V_E] = \sqrt{V_E^2 + H^2[V_E]}$. Likewise, the phase angle of the analytic signal is defined as $Arg[V_E] = \arctan\big[H[V_E]/V_E\big]$. Analogously, we define $Env[LFP]$ and $Arg[LFP]$ to estimate the amplitude and phase of the envelope of the experimental LFP.

The transition between noise-induced and self-sustained oscillations happens when the real part of the eigenvalue is zero. This condition is expressed as

$$\lambda_0 = -\frac{\alpha_E E_0}{2}\left[1 - \frac{\alpha_E E_0}{(1-E_0)\beta_E}\right] W_{EE} + \frac{\alpha_I I_0}{2}\left[1 - \frac{\alpha_I I_0}{(1-I_0)\beta_I}\right] W_{II} + \frac{\alpha_E}{2(1-E_0)} + \frac{\alpha_I}{2(1-I_0)} = 0\,. \tag{49}$$

Therefore the expressions of a, b and c in Eq.11 are given as:

$$a = \frac{\alpha_E E_0}{2}\left[1 - \frac{\alpha_E E_0}{(1-E_0)\beta_E}\right], \quad b = \frac{\alpha_I I_0}{2}\left[1 - \frac{\alpha_I I_0}{(1-I_0)\beta_I}\right], \quad and \quad c = \frac{\alpha_E}{2(1-E_0)} + \frac{\alpha_I}{2(1-I_0)}$$

This analytical expression is in good agreement with numerical simulations (Fig. 7**(B-Left)**). The stability boundary is shown as dots in black, and is reproduced in Figs. 7**(B-Right)**. We also computed the amplitude ratio, phase difference and excitatory and inhibitory frequency distributions from the simulated linear stochastic Eqs. (26-27). The results show approximate Gaussian distributions around the mean values provided by the linear stability analysis in Eqs. (46-48), in spite of the linearities in Eqs. (26-27). The noise basically spreads the delta distributions of the deterministic case to yield approximately Gaussian distributions. The standard deviations of the distributions increase with noise. Numerically computed means of the frequency content are close to the imaginary part of the complex conjugate eigenvalues since the system is close to the Hopf bifurcation (black dot in Figs.7). Moving away from this boundary slowly diminishes the agreement as the frequency depends on the amplitude.

Envelope-phase decomposition of the Linear SWC. By taking into account the amplitude ratio and phase difference between the E and I dynamics derived from the linear analysis, one can define the E and I fluctuations in the following form[9]:

$$V_E = Z_E \cos\left(\omega_0 t + \phi_E\right); \quad V_I = \alpha Z_E \cos\left(\omega_0 t + \phi_E + \delta\right) \tag{50}$$

where Z_E and ϕ_E represent the stochastic amplitude and phase of the E fluctuations. The idea is to compute the dynamics of the slow amplitude fluctuations of the cycle to cycle amplitude, Z_E, and the phase ϕ_E of the E fluctuations (see [7, 9]). The parameters α, δ and ω_0 are the amplitude ratio, phase difference and frequency computed from the linear stability analysis Eqs. (46-48). The stochastic averaging method [9] then yields:

$$dZ_{E(t)} = \left[-\lambda_0 Z_E + \frac{D}{2Z_E}\right] dt + \sqrt{D} dW_1(t) \tag{51}$$

$$d\phi_E(t) = \frac{\sqrt{D}}{Z_E} dW_2(t) \tag{52}$$

with the effective noise strength:

$$D = \frac{(\sigma_I)^2 + (\alpha\sigma_E)^2}{2\alpha^2 \sin^2(\delta)} \,. \tag{53}$$

Here $W_1(t)$ and $W_2(t)$ are two independent Wiener processes. Further, since the noise is only additive, there is no need to distinguish between Itô and Stratonovich calculus; the SAM method itself was developed using Itô calculus [37, 38].

When the noises on the E and I populations are identical ($\sigma_E = \sigma_I = \sigma$), the expression of D can be reformulated following the calculations below:

$$\begin{aligned} D &= \frac{(\sigma_I)^2 + (\alpha\sigma_E)^2}{2\alpha^2 \sin^2(\delta)} \\ &= \frac{(1+\alpha^2)\sigma^2}{2\alpha^2 \sin^2(\delta)} \\ &= \frac{1}{2\sin^2(\delta)}\left(1 + \frac{(1-E_0)\beta_E f^{'}(s_{E_0})}{(1-I_0)\beta_I f'(s_{I_0})}\frac{W_{EI}}{W_{IE}}\right)\sigma^2 \\ &= \frac{1}{2\sin^2(\delta)}\left(1 + \gamma\frac{W_{EI}}{W_{IE}}\right)\sigma^2. \end{aligned}$$

Thus,

$$D = \frac{1}{2\sin^2(\delta)}\left(1 + \gamma\frac{W_{EI}}{W_{IE}}\right)\sigma^2$$

with

$$\gamma \equiv \frac{(1-E_0)\beta_E f^{'}(s_{E_0})}{(1-I_0)\beta_I f'(s_{I_0})} = \frac{\alpha_E E_0\left[1 - \dfrac{\alpha_E E_0}{(1-E_0)\beta_E}\right]}{\alpha_I I_0\left[1 - \dfrac{\alpha_I I_0}{(1-I_0)\beta_I}\right]} = \frac{a}{b} \,.$$

Stochastic analysis of the amplitude dynamics

To perform the stochastic analysis of the amplitude process, we consider the corresponding stationary Fokker-Planck equation of (51) [39]:

$$-\frac{d}{dZ_E}\left[\left(-\lambda_0 Z_E + \frac{D}{2Z_E}\right)P_E\right] + \frac{D}{2}\frac{d^2 P_E}{dZ_E^2} = 0 \tag{54}$$

where $P_E\big(Z_E\big)$ represents the stationary probability density function of the amplitude process $Z_E(t)$. We define the following parameters

The normalized solution $\left(\int_0^\infty P_E(Z_E) = 1\right)$ of this second order differential equation is given by

$$P_E(Z_E) = \frac{2\lambda_0}{D} Z_E \exp\left(-\frac{\lambda_0}{D} Z_E^2\right). \tag{55}$$

By using the relation between the amplitude of excitatory and inhibitory LFPs ($Z_I = \alpha Z_E$), we obtain the stationary probability density function for the inhibitory amplitude process:

$$P_I(Z_I) = \frac{2\alpha^2\lambda_0}{D} Z_I \exp\left(- \frac{\alpha^2\lambda_0}{D} Z_I^2 \right). \tag{56}$$

where the peaks of the stationary densities, which represents the most probable values taken by the stochastic amplitude process, are respectively given by

$$Z_E^* = \sqrt{\frac{D}{2\lambda_0}}\,, \qquad Z_I^* = \alpha\sqrt{\frac{D}{2\lambda_0}}\,. \tag{57}$$

Theory of gamma rhythms Local Field potentials in the monkey V1 area

Here we use a more general approach to deriving LFP dynamics from the E-I equations that assumes the LFP is the superposition of both E and I components:

$$V(t) = a_1 V_E(t) + a_2 V_I(t), \quad 0 \le a_1, a_2 \le 1. \tag{58}$$

This equation can be reduced as follows

$$\begin{aligned} V(t) &= a_1 V_E(t) + a_2 V_I(t) \\ &= a_1 Z_E(t)\cos\big(\omega_0 t + \phi_E(t)\big) + a_2\alpha Z_E(t)\cos\big(\omega_0 t + \phi_E(t) + \delta\big) \\ &= a_1 Z_E(t)\cos\big(\omega_0 t + \phi_E(t)\big) + a_2\alpha Z_E(t)\Big[\cos(\omega_0 t + \phi_E(t))\cos(\delta) - \sin(\omega_0 t + \phi_E(t))\sin(\delta)\Big] \\ &= Z_E(t)\Big[\big(a_1 + a_2\alpha\cos(\delta)\big)\cos(\omega_0 t + \phi_E(t)) - a_2\alpha\sin(\delta)\sin(\omega_0 t + \phi_E(t))\Big]. \end{aligned}$$

We further assume that a_1 and a_2 are chosen such that

$$\cos(\beta) = a_1 + a_2\alpha\cos(\delta), \quad \sin(\beta) = a_2\alpha\sin(\delta) \quad \textit{and} \quad \beta = \arctan\left(\frac{a_2\alpha\sin(\delta)}{a_1 + a_2\alpha\cos(\delta)}\right).$$

Then,

$$\begin{aligned} V(t) &= Z_E(t)\Big[\cos(\beta)\cos(\omega_0 t + \phi_E(t)) - \sin(\beta)\sin(\omega_0 t + \phi_E(t))\Big] \\ &= Z_E(t)\cos(\omega_0 t + \phi_E(t) + \beta) \end{aligned}$$

We make the following changes of variables:

$$Z(t) = Z_E(t) \qquad \phi(t) = \phi_E(t) + \beta\,.$$

Therefore the LFP can be defined as

$$V(t) = Z(t)\cos\big(\omega_0 t + \phi(t)\big)\,. \tag{59}$$

The envelope $Z(t)$ and the phase $\phi(t)$ are then described as follows:

$$dZ(t) = \Big[-\lambda_0 Z(t) + \frac{D}{2Z(t)}\Big]\,dt + \sqrt{D}dW_1(t) \tag{60}$$

$$d\phi(t) = \frac{\sqrt{D}}{Z(t)}dW_2(t) \tag{61}$$

with the corresponding pdf

$$P(Z) = \theta Z \exp\Big(-\theta\frac{Z^2}{2}\Big) \tag{62}$$

with

$$\theta = \frac{2\lambda_0}{D}\,. \tag{63}$$

The problem of building a theory for LFP data in the context of noise-induced rhythms reduces to that of estimating the three parameters λ_0, ω_0 and D such that the synthetic data generated by the Eqs. 59-61 have the same statistics (envelope, LFP, frequency densities, ...) than the real data. We now describe how to extract these parameters.

Estimating λ_0 from the data

From the observation of the autocorrelation function of the LFP data, we can see a decay after a few cycles. Is there some relation between such decay and the value of λ_0 and ω_0? To answer this question, let find theoretically if similar decay exists in the Linear SWC. Let first put Eqs.26-27 in the following compact form:

$$\frac{d\mathbf{V(t)}}{dt} = \mathbf{A}\mathbf{V(t)} + \sigma\xi(\mathbf{t}), \tag{64}$$

where:

$$\mathbf{V(t)} = \begin{pmatrix} V_E(t) \\ V_I(t) \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} A_{EE} & A_{EI} \\ A_{IE} & A_{II} \end{pmatrix}, \quad \sigma = \begin{pmatrix} \sigma_E & 0 \\ 0 & \sigma_I \end{pmatrix} \quad \text{and} \quad \xi(\mathbf{t}) = \begin{pmatrix} \xi_E(t) \\ \xi_I(t) \end{pmatrix}.$$

We compute the time correlation matrix of the 2D linear SWC using the stationary assumption as follows [39]:

$$\langle \mathbf{V_s}(\mathbf{t}), \mathbf{V_s^T}(\mathbf{u})\rangle = \exp\Big[\mathbf{A}\big(t-u\big)\Big]\int_{-\infty}^{u}\exp\Big[\mathbf{A}\big(u-t^{'}\big)\Big]\sigma\sigma^{\mathbf{T}}\exp\Big[\mathbf{A^T}\big(u-t^{'}\big)\Big]dt^{'} \tag{65}$$

$$= \exp\Big[\mathbf{A}\big(t-u\big)\Big]\mathbf{\Sigma} \qquad t > u, \tag{66}$$

$$= \mathbf{\Sigma}\exp\Big[\mathbf{A^T}\big(u-t\big)\Big] \qquad t < u. \tag{67}$$

The subscript s stands for stationary and the superscript T refers to the transpose of a given matrix. Furthermore Σ is the stationary covariance matrix defined as

$$\Sigma = \frac{-\big(DetA\big)\sigma\sigma^T - \big[-A + (TrA)I_2\big]\sigma\sigma^T\big[-(TrA)I_2 + A\big]^T}{2(TrA)(DetA)}. \tag{68}$$

Since the autocorrelation function is an even function, we focus only on the case $\tau = t - u > 0$:

$$\langle \mathbf{V_s}(\mathbf{t}), \mathbf{V_s^T}(\mathbf{t}-\tau)\rangle = \exp\big[A\tau\big]\Sigma. \tag{69}$$

The matrix $\mathbf{A}$ diagonalized as

$$\mathbf{A} = \mathbf{PDP^{-1}} \tag{70}$$

where $\mathbf{P}$ is the transition matrix, $\mathbf{P^{-1}}$ its inverse and $\mathbf{D}$ the corresponding diagonal matrix (with complex conjugate eigenvalues in the diagonal and zero elsewhere) of $\mathbf{A}$. This leads to

$$\mathbf{A}\tau = \mathbf{P}(\mathbf{D}\tau)\mathbf{P^{-1}}. \tag{71}$$

Thus,

$$\begin{aligned}
\exp\big[\mathbf{A}\tau\big] &= \mathbf{P}\exp\big[\mathbf{D}\tau\big]\mathbf{P^{-1}} && (72)\\
&= \mathbf{P}\begin{pmatrix} e^{(-\lambda_0 + j\omega_0)\tau} & 0 \\ 0 & e^{(-\lambda_0 - j\omega_0)\tau}\end{pmatrix}\mathbf{P^{-1}} && (73)\\
&= \mathbf{P}\begin{pmatrix} e^{j\omega_0\tau} & 0 \\ 0 & e^{-j\omega_0\tau}\end{pmatrix}\mathbf{P^{-1}}\exp(-\lambda_0\tau). && (74)
\end{aligned}$$

Finally,

$$\langle \mathbf{V_s}(\mathbf{t}), \mathbf{V_s^T}(\mathbf{t}-\tau)\rangle = \left[\mathbf{P}\begin{pmatrix} e^{j\omega_0\tau} & 0 \\ 0 & e^{-j\omega_0\tau}\end{pmatrix}\mathbf{P^{-1}}\mathbf{\Sigma}\right]\exp(-\lambda_0\tau). \tag{75}$$

By solving the eigenvector equations $\mathbf{A}\mathbf{v}_\pm = (-\lambda_0 \pm j\omega_0)\mathbf{v}_\pm$, we obtain the following expressions:

$$\mathbf{v}_+ = \begin{pmatrix} 1 \\ -\frac{A_{EE}+\lambda_0}{A_{EI}} + j\frac{\omega_0}{A_{EI}}\end{pmatrix} \quad \text{and} \quad \mathbf{v}_- = \begin{pmatrix} 1 \\ -\frac{A_{EE}+\lambda_0}{A_{EI}} - j\frac{\omega_0}{A_{EI}}\end{pmatrix}.$$

Thus the matrix $\mathbf{P}$ and its inverse $\mathbf{P^{-1}}$ are given respectively by

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -\frac{A_{EE}+\lambda_0}{A_{EI}} + j\frac{\omega_0}{A_{EI}} & -\frac{A_{EE}+\lambda_0}{A_{EI}} - j\frac{\omega_0}{A_{EI}}\end{pmatrix}, \quad \text{and} \quad \mathbf{P^{-1}} = \frac{A_{EI}}{-2j\omega_0}\begin{pmatrix} -\frac{A_{EE}+\lambda_0}{A_{EI}} - j\frac{\omega_0}{A_{EI}} & -1 \\ \frac{A_{EE}+\lambda_0}{A_{EI}} - j\frac{\omega_0}{A_{EI}} & 1\end{pmatrix}.$$

The expansion of Eq.75 leads to the following coefficients of the Correlation matrix

$$\langle V_E(t), V_E(t-\tau)\rangle = \left[\Sigma_{11}\cos(\omega_0\tau) + \left(\left(\frac{A_{EE}+\lambda_0}{\omega_0}\right)\Sigma_{11} + \frac{A_{EI}}{\omega_0}\Sigma_{21}\right)\sin(\omega_0\tau)\right]\exp(-\lambda_0\tau) \tag{76}$$

$$\langle V_E(t), V_I(t-\tau)\rangle = \left[\Sigma_{12}\cos(\omega_0\tau) + \left(\left(\frac{A_{EE}+\lambda_0}{\omega_0}\right)\Sigma_{12} + \frac{A_{EI}}{\omega_0}\Sigma_{22}\right)\sin(\omega_0\tau)\right]\exp(-\lambda_0\tau) \tag{77}$$

$$\langle V_I(t), V_E(t-\tau)\rangle = \left[\Sigma_{21}\cos(\omega_0\tau) + \left(\frac{A_{IE}}{\omega_0}\Sigma_{11} - \left(\frac{A_{EE}+\lambda_0}{\omega_0}\right)\Sigma_{21}\right)\sin(\omega_0\tau)\right]\exp(-\lambda_0\tau) \tag{78}$$

$$\langle V_I(t), V_I(t-\tau)\rangle = \left[\Sigma_{22}\cos(\omega_0\tau) + \left(\frac{A_{IE}}{\omega_0}\Sigma_{12} - \left(\frac{A_{EE}+\lambda_0}{\omega_0}\right)\Sigma_{22}\right)\sin(\omega_0\tau)\right]\exp(-\lambda_0\tau). \tag{79}$$

The $\Sigma_{i,j}$ are the coefficients of the covariance matrix, and have the following expressions:

$$\Sigma_{11} = \frac{\sigma_E^2}{4\lambda_0} + \frac{A_{II}^2\sigma_E^2 + A_{IE}^2\sigma_I^2}{4\lambda_0(\lambda_0^2+\omega_0^2)} \tag{80}$$

$$\Sigma_{12} = \Sigma_{21} = -\frac{(A_{II}A_{IE}\sigma_E^2 + A_{EE}A_{EI}\sigma_I^2)}{4\lambda_0(\lambda_0^2+\omega_0^2)} \tag{81}$$

$$\Sigma_{22} = \frac{\sigma_I^2}{4\lambda_0} + \frac{A_{IE}^2\sigma_E^2 + A_{EE}^2\sigma_I^2}{4\lambda_0(\lambda_0^2+\omega_0^2)}. \tag{82}$$

We can compute the correlation function of the LFP defined in Eq.58 as (do not confuse with the vector formulation in bold):

$$\begin{aligned}\big\langle V(t), V(t-\tau)\big\rangle &= \left\langle \Big(a_1V_E(t) + a_2V_I(t)\Big), \Big(a_1V_E(t-\tau) + a_2V_I(t-\tau)\Big)\right\rangle \\ &= a_1^2\Big\langle V_E(t), V_E(t-\tau)\Big\rangle + a_1a_2\Big\langle V_E(t), V_I(t-\tau)\Big\rangle \\ &+ a_1a_2\Big\langle V_I(t), V_E(t-\tau)\Big\rangle + a_2^2\Big\langle V_I(t), V_I(t-\tau)\Big\rangle.\end{aligned}$$

Using the correlation matrix coefficients defined in Eqs.76-79, we arrive at

$$\big\langle V(t), V(t-\tau)\big\rangle = \big[A\cos(\omega_0\tau) + B\sin(\omega_0\tau)\big]\exp(-\lambda_0\tau) \tag{83}$$

with

$$A = a_1^2\Sigma_{11} + 2a_1a_2\Sigma_{12} + a_2^2\Sigma_{22} \tag{84}$$

$$B = \left(\frac{A_{EE}+\lambda_0}{\omega_0}\right)\left(a_1^2\Sigma_{11} - a_2^2\Sigma_{22}\right) + \frac{A_{EI}}{\omega_0}\left(a_1^2\Sigma_{21} + a_1a_2\Sigma_{22}\right) + \frac{A_{IE}}{\omega_0}\left(a_1a_2\Sigma_{11} + a_2^2\Sigma_{12}\right). \tag{85}$$

The normalized autocorrelation function is then obtained as

$$C(\tau) = \frac{\big\langle V(t), V(t-\tau)\big\rangle}{\big\langle V^2(t)\big\rangle} = \big[\cos(\omega_0\tau) + C_0\sin(\omega_0\tau)\big]\exp(-\lambda_0\tau) \tag{86}$$

with

$$C_0 = \frac{B}{A} = \frac{\Big(A_{EE} + \lambda_0\Big)\Big(a_1^2\Sigma_{11} - a_2^2\Sigma_{22}\Big) + A_{EI}\Big(a_1^2\Sigma_{21} + a_1a_2\Sigma_{22}\Big) + A_{IE}\Big(a_1a_2\Sigma_{11} + a_2^2\Sigma_{12}\Big)}{\omega_0\big(a_1^2\Sigma_{11} + 2a_1a_2\Sigma_{12} + a_2^2\Sigma_{22}\big)}. \tag{87}$$

Following Eqs.86-87, the normalized autocorrelation function of $V(t)$ is an oscillatory function that decays exponentially with a rate λ_0. Therefore, to extract λ_0 from the data, it suffices to compute the corresponding autocorrelation, to identify its decay and to fit such decay with the function $e^{-\lambda_0\tau}$. The autocorrelation function of the data allows us to extract the value of λ_0. We compute the peaks of the normalized autocorrelation function at multiples of the period T_0 ($\tau = kT_0$):

$$C(kT_0) = \exp\big(-kT_0\lambda_0\big), \quad k = 0, 1, 2, ...M-1. \tag{88}$$

The cost function is:

$$K = \frac{1}{2M}\sum_{k=0}^{M-1}\Big[\exp\big(-kT_0\lambda_0\big) - \tilde{C}(kT_0)\Big]^2 /, , \tag{89}$$

where we focus on 5 peaks including the one at lag zero, i.e. we set $M = 5$. Using gradient descent, we obtain the following recurrence relation for the value of λ_0 after $i+1$ iterations:

$$\lambda_0^{i+1} = \lambda_0^i - \frac{\alpha}{M}\sum_{k=0}^{M-1}\Big(-kT_0\lambda_0^i\exp\big(-kT_0\lambda_0^i\big)\Big)\Big[\exp\big(-kT_0\lambda_0^i\big) - \tilde{C}(kT_0)\Big]. \tag{90}$$

The expression $\tilde{C}(kT_0)$ represents the value of the normalized autocorrelation function at each peak extracted from the data and kT_0 the corresponding location. The optimal value of λ_0 is obtained when the cost function reaches a global minimum and the value $\lambda_0^{i+1} = \lambda_0^i$. The hyperparameter α is the learning rate and M the number of autocorrelation maxima used.

Estimating the mean angular frequency ω_0. The value of ω_0 is the angular frequency of the autocorrelation function Eq.86 and is defined as

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T_0} \tag{91}$$

f_0 and T_0 are respectively the mean frequency and the mean period. The values of T_0 can be easily obtained as the location of the first peak after the peak at the origin of the autocorrelation function. The parameter ω_0 may also be inferred through the location of the peak of the power spectrum. However, the peak location does not match with the mean frequency when the power spectrum is not Gaussian. This can lead to different but close results between the two inference approaches. Here, we chose the first approach.

Estimating the effective noise intensity D: Probabilistic Inference. Each trial consists in the presentation of the same stimulus during trials 2s, plus a short period before the stimulus onset and a much longer epoch after

the stimulus onset. Here, we only consider the period after the stimulus onset. We focus on the likelihood that the contrast is encoded in the parameter θ which parametrizes the envelope response $P(Z|\theta)$. The likelihood function can be defined as

$$P(Z|\theta) = \theta Z \exp\Big(- \theta \frac{Z^2}{2}\Big). \tag{92}$$

To infer the value of D, it is obviously appropriate to first infer the value of $\hat{\theta}$ that maximizes the Likelihood. As usually we proceed by finding the parameter $\hat{\theta}$ that maximizes the log-likelihood:

$$\hat{\theta} = \arg\max_{\theta} \ \log P(Z|\theta)\,. \tag{93}$$

The value of D can therefore be obtained by considering that $\hat{\theta}$ is equal to the value in Eq.63. The value of $\hat{\theta}$ can be derived as follows:

$$\begin{aligned} L(\theta) = P(Z|\theta) &= \theta Z \exp\Big(- \theta \frac{Z^2}{2}\Big), \\ &= \prod_{j=1}^{n} P(Z_j|\theta) = \prod_{j=1}^{n} \theta Z_j \exp\Big(- \theta \frac{Z_j^2}{2}\Big), \end{aligned} \tag{94}$$

The log-likelihood is obtained as

$$\begin{aligned} \log L(\theta) &= \log\Big(\prod_{j=1}^{n} \theta Z_j \exp\Big(- \theta \frac{Z_j^2}{2}\Big)\Big) \\ &= \sum_{j=1}^{n} \log\Big(\theta Z_j \exp\Big(- \theta \frac{Z_j^2}{2}\Big)\Big) \\ &= \sum_{j=1}^{n} \Big[\log\Big(\theta Z_j \Big) - \theta \frac{Z_j^2}{2}\Big] \\ &= \sum_{j=1}^{n} \Big[\log\big(Z_j \big) + \log(\theta) - \theta \frac{Z_j^2}{2}\Big]\,. \end{aligned} \tag{95}$$

For simplicity, let us set

$$\big\langle \log(Z) \big\rangle = \frac{1}{n}\sum_{j=1}^{n} \log(Z_j), \quad \big\langle Z^2 \big\rangle = \frac{1}{n}\sum_{j=1}^{n} Z_j^2\,.$$

The log-likelihod function can be simplified to:

$$\log L(\theta) = n\big\langle \log(Z) \big\rangle + n\log(\theta) - n\frac{\theta}{2}\big\langle Z^2 \big\rangle\,. \tag{96}$$

The maximum log-likelihood estimate is obtained by taking the derivative of Eq.96 equal to zero,

$$\frac{d\log L(\theta)}{d\theta} = 0\,, \tag{97}$$

leading to the estimate

$$\hat{\theta} = \frac{2}{\langle Z^2 \rangle} = \frac{2\lambda_0}{D}\,.$$

The value of D can therefore be obtained as

$$D = \lambda_0 \langle Z^2 \rangle\,. \tag{98}$$

Fisher Information

The Fisher information is given by the following relation

$$I_F(\hat{\theta}) = \left\langle \left(\frac{\partial^2 \log P(\tilde{\mathbf{Z}}|\theta)}{\partial^2 \theta} \right) \right\rangle_{\hat{\theta}}.$$

A straightforward result is obtained as $\frac{n\left(\langle \tilde{\mathbf{Z}}^2 \rangle\right)^2}{4}$ after using Eq. 96. Where n is the number of samples of the LFP considered.

Data Analysis

We consider previous datasets recorded from the M1 monkey's primary visual cortex during a fixation task (data obtained online from Peter de Weerd's Lab). The recording procedures and the exact details of the task are described in previous works. A given dataset consists of several trials of a few seconds each, recorded at the same value of the stimulus contrast. Precisely, the datasets are recorded for different values of the stimulus contrasts ranging from values of $2, 9, 16.3, 35.9, 50.3$ to 72. However, we performed the analysis on only the four last datasests, they are those that show a clear peak of their power spectra in the gamma band. For these datasests, each trial is composed of a short pre-stimulus and a much longer stimulated epochs. For each trial, we discarded the pre-stimulus epoch and an epoch of 100 ms after the stimulus onset. This allows to focus our analysis only on the epochs where the response to the stimulus is stationary and to avoid the effects of the transients that follow the stimulus onset. To build a single timeserie dataset, we concatenated all the stationary stimulated epochs previously extracted. Repeating the procedure for the four stimulus contrasts allows to obtain four timeserie datasets that are analyzed throughout the paper. Furthermore, we normalized each timeserie dataset using a z-score procedure. We are interested only in the gamma band activity, thus we applied a band-pass filter to each of the timeserie. The limits of each band-pass filter are chosen to obtain a power spectrum of the timeserie that is as close as possible to the unfiltered timeserie power spectrum in the corresponding gamma band, and close as possible to a Gaussian shape. To compute the envelope processes used to perform the inference procedure, we applied the Hilbert transform on each of the filtered timeseries. The values of the coefficients λ_0 and ω_0 are obtained directly from the filtered timeseries after computing their autocorrelation functions. The fitted parameters λ_0, ω_0 and D are used to construct the envelope-phase decom-

position and the corresponding Local Field potential timeseries. The envelope-phase decomposition and their Local field potentials timeseries are obtained from numerical simulation using the Euler-Maruyama scheme. The timestep of $dt = 1/Fs = 0.001s$ was used during the simulation, where $Fs = 1000Hz$ is the sampling rate used during the recording of the LFPs. The simulated envelopes are directly compared to the envelopes obtained from the Hilbert transform applied on the filtered datasets.

Burst definition and identification

Gamma bursts refer to "short" and random epochs of high gamma power. They can be identified in the spectrograms (Fig.4**(A)**) as epochs in red. Here, we formally define a burst as the epoch the envelope process spends above a given threshold (Fig.4**(A)**).

Burst features

We further define three burst features [3, 40, 41]:

:Burst duration: the duration the envelope spends above the threshold during a burst (Fig.4**(A)**). Burst amplitude : the maximal amplitude of the envelope above the threshold during a burst (Fig.4**(A)**). Burst frequency: the number of oscillation cycles during a burst divided by the burst duration.

Burst threshold. From our burst definition, it follows that burst feature values depend on the threshold. A lower threshold will lead to more bursts, but many of them will not be seen in the spectrogram. A higher threshold will lead to few bursts and we will miss some bursts seen in the spectrogram. Here we chose a value of $th = 0.2$ which allows most of the bursts to be visible in the spectrogram for the different contrasts.

Burst classification. Following the fact that our burst identification procedure detects bursts that are very shorts and not seen in the spectrogram, we divided the bursts into two classes:

- The True Bursts (TB): the envelope process spends at least three oscillation cycles above the threshold. They were almost all seen in the spectrogram (Fig.4-(**(A)**). They are considered more prominent during stimulation.
- The False Bursts (FB): the envelope process spends less than three oscillation cycles above the threshold. Most of them were not seen in the spectrogram. However, few of them occurring during brief random high amplitude excursions were observed in the spectrogram. They are more associated with noise. The bursts with no oscillation cycle above the threshold were discarded and not considered during burst feature analysis.

Technically, we define a burst as the epoch the gamma rhythms envelope spends above a given threshold [40, 41]. We characterize each burst by its features like the amplitude, duration and frequency [40, 41]. We perform burst-feature analysis for both the data and the constructed envelope-phase model, and this for the four stimulus conditions. Our results show good agreements between the distributions and the mean values of the burst features from the data and the constructed envelope-phase model. We found that the values of the mean burst duration, amplitude and frequency are shaped by the stimulus contrast. For all stimulus conditions, there was a positive correlation between the burst durations and amplitudes, and a negative correlation between their frequencies and amplitudes, meaning that high

amplitude bursts have longer durations and high frequency bursts have shorter amplitudes. In addition, gamma rhythm responses with longer burst durations and higher burst amplitudes showed optimal information encoding, specified by higher values of the FI. In contrast, the response with shorter burst durations was associated to a bad stimulus encoding through a weak value of the FI. Interestingly, higher burst amplitudes were required but not critical to achieve an optimal information coding. A response with higher burst amplitudes could have a lower FI value if the corresponding burst durations were substantially shorts. Longer burst durations are critical to achieve optimal information encoding. Therefore, burst features can be used as markers of stimulus encoding.

For the analysis of gamma bursts, the LFPs obtained through the envelope-phase decomposition were filtered to remove fast and noisy fluctuations due to the simulation process and to have a better comparison with the filtered datasets. For the datasets and their corresponding envelope-phase decomposition, the bursts were extracted using the burst definition and the procedure described in the main text. Precisely, a burst a defined as the epoch the envelope process spends above a specific threshold. The threshold was set at the value $th = 0.2$ for all the four datasets and their corresponding envelope-phase decomposition. The bursts were labeled as True or False if their corresponding number of cycles above the threshold was greater of less than 3 respectively. We performed the analysis separately for False and True bursts and compared the results of each of the two classes between the data and the envelope-phase model. The analysis consists on extracting the means, the standard deviations, the marginal distributions of the burst duration, amplitude and frequency, as well as the joint distributions between burst duration and amplitude, and burst amplitude and frequency. During the analysis, we only considered bursts with certain properties. Specifically, we only took into account burst with duration and amplitude less than particular values. We chose these criteria to remove outliers in our analysis and to allow a better comparison of the mean and standard deviation.

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